The maximum number of cycles in a graph of large rank

The formulas of Horak and Kim are exact for every large rank

KNT · Astra · Fable 5.2

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Let $\psi(r)$ be the largest number of cycles in a graph of cyclomatic number r. We prove that, for all sufficiently large r,

$$\psi(2q+1) = 2^{2q} + \left(2q + \tfrac{1}{2}\right)2^q - 3q, \qquad \psi(2q+2) = 2^{2q+1} + \left(2q + \tfrac{9}{2}\right)2^q - 3q - 4.$$

Cubic graphs of Horak and Kim have these numbers of cycles. In particular $\psi(r) \sim 2^{r-1}$, which Entringer and Slater conjectured in 1981. Astra and Fable 5.2 are language models. Section 1.4 states the part of each author. The theorem has a formal proof in Lean 4, and the Lean kernel has checked it.

The proof in eight sentences

1. Subdividing an edge keeps the number of cycles, and splitting a vertex of degree four or more does not lower it. So some cubic multigraph G of rank r has $\psi(r)$ cycles.
2. Fix a short cycle W. A cycle through an edge of W is determined by its part outside W, and this part is a forest. By the *forest estimate*, at most $\frac{1}{2} + o(1)$ of all outside parts are forests, so $\psi(r) \le (\frac{1}{2} + o(1))2^r$.
3. A cut with at most four edges divides G into two parts, and a cycle across the cut consists of paths in the parts. A part with more than half of the possible paths between two ends, by a fixed margin, is a ladder up to a bounded error, and it has few cycles.
4. So a graph with such a cut is a ring of copies of $K_{2,2}$ and of single edges, or it has at most $2^{r-1} + O(2^{r/2})$ cycles.
5. A ring has at most $H(r)$ cycles, by a direct count; $H(r)$ is the right side of the formulas above.
6. Without such a cut but with a short cycle, G has fewer than 2^{r-1} cycles: sort the cycles through an edge of a shortest cycle by the vertex where they leave it.
7. Without such a cut and with large girth, G has fewer than 2^{r-1} cycles: by a second-moment argument, G contains two large pieces with small boundaries, and each piece has a forest proportion of at most $\frac{3}{5}$.
8. $H(r) - 2^{r-1}$ is at least $(r-2)2^{r/2-1}$. So $\psi(r) = H(r)$ for large r.

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1 The result and the plan

1.1 The result

Throughout, $\log$ is the binary logarithm, and $\log^{(j)}$ is its j-fold iterate. For a graph G, an *even subgraph* is a set of edges that meets every vertex an even number of times. With symmetric difference as addition, the even subgraphs form a vector space $\mathcal{Z}(G)$ over $\mathbb{F}_2$. Its dimension is

$$\beta(G) = |E(G)| - |V(G)| + \operatorname{comp}(G),$$

where $\operatorname{comp}(G)$ is the number of components. We call $\beta(G)$ the *rank* of G; other names are cyclomatic number and cycle rank. A *cycle* is a connected 2-regular subgraph. We identify a subgraph without isolated vertices with its edge set. Then a cycle is a nonzero element of $\mathcal{Z}(G)$, so a graph of rank r has at most $2^r - 1$ cycles.

Let $\psi(r)$ be the largest number of cycles in a graph of rank r.

- Entringer and Slater [4] proved that $2^{r-1} < \psi(r) < 2^r$ for $r \geq 3$, and they conjectured that $\psi(r) \sim 2^{r-1}$.
- Aldred and Thomassen [2] proved that $\psi(r) \leq \frac{15}{16}2^r$.
- Alahmadi, Aldred, dela Cruz, Solé and Thomassen [1] proved that a connected graph with p vertices and more than $2p + 3\log(3p)$ edges has at most 2^{r-1} cycles.
- Horak and Kim [5] found cubic graphs of rank r with $H(r)$ cycles, where

$$H(2q+1) = 2^{2q} + \left(2q + \tfrac{1}{2}\right)2^q - 3q, \qquad H(2q+2) = 2^{2q+1} + \left(2q + \tfrac{9}{2}\right)2^q - 3q - 4. \tag{1.1}$$

For odd rank $r = 2q + 1$ the graph of Horak and Kim is a ring of q copies of $K_{2,2}$, each joined to the next by two edges. For even rank, one more edge is inserted across the ring (Figure 1). We call this graph R_r.

Theorem 1.1 (the maximum number of cycles)

There is an integer R such that $\psi(r) = H(r)$ for every $r \geq R$.

In particular $\psi(r) = 2^{r-1} + (1 + o(1))\, r\, 2^{\lfloor (r-1)/2 \rfloor}$, so $\psi(r) \sim 2^{r-1}$. The proof gives no value for R.

> **Fine print.** The theorem is false for small r. For example, take the vertices u_i, v_i with $i \in \mathbb{Z}/26$ and the edges u_iu_{i+1}, u_iv_i and v_iv_{i+10}. This generalised Petersen graph has rank 27, and a direct count by computer gives 67 959 973 cycles, while $H(27) = 67\,325\,913$.

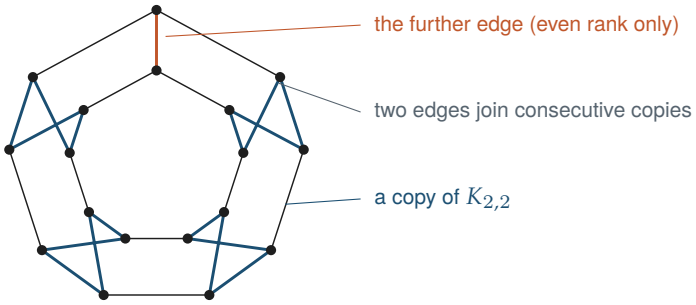


Figure 1. The graph R_{10}: a ring of four copies of $K_{2,2}$ and one further edge. It has rank 10 and $H(10) = 696$ cycles.

1.2 The five cases of the proof

The number of cycles does not change when we delete a vertex of degree at most one, or replace a vertex of degree two and its two edges by one edge. A vertex of degree at least four can be split into vertices of degree three, and no cycle is lost. So it suffices to count the cycles of a connected multigraph G of rank r in which all vertices have degree three (Lemma 3.10). The graphs R_r show that $\psi(r) \geq H(r)$.

For the upper bound there are five cases (Figure 2). A *cyclic cut* is a set of edges whose deletion leaves two parts that each contain a cycle. The number $H(r) - 2^{r-1}$ is of order $r\,2^{r/2}$, so the five bounds give the theorem (Section 9).

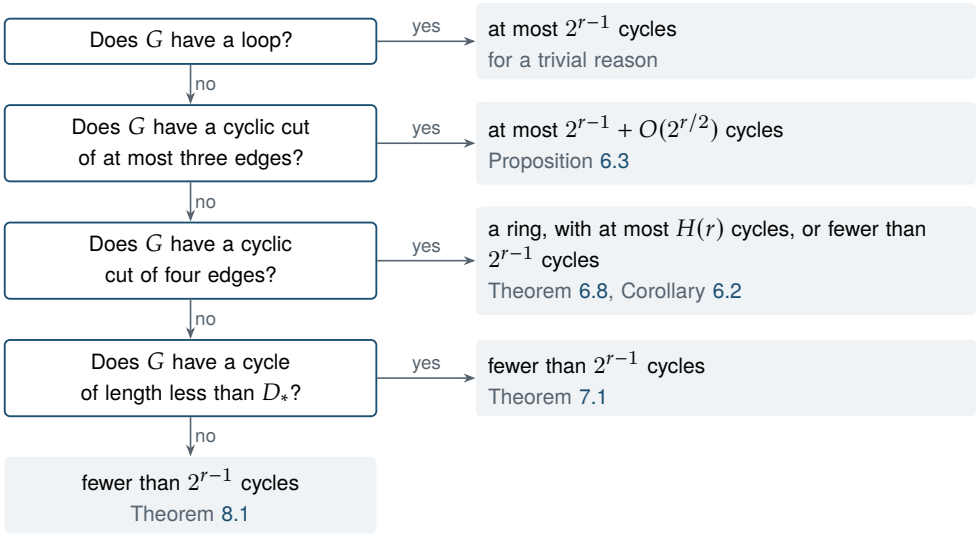


Figure 2. The five cases for a connected cubic multigraph G of large rank r. The number D_* is an absolute constant.

1.3 Outside parts and the forest estimate

Fix a subgraph W of G. For an even subgraph ξ we call $\xi \setminus E(W)$ its *outside part* (Figure 3). If a cycle uses an edge of W, then its outside part is a proper subset of a cycle, so it is a forest.

IDEA | cycles and outside parts

Let W be a cycle and e an edge of W. Different cycles through e have different outside parts, and each of these outside parts is a forest (Lemma 2.1). By the *forest estimate* (Theorem 3.1), at most $\frac{1}{2} + o(1)$ of all outside parts are forests when W is small. An edge recursion then gives $\psi(r) \le (\frac{1}{2} + o(1))2^r$ (Corollary 3.11).

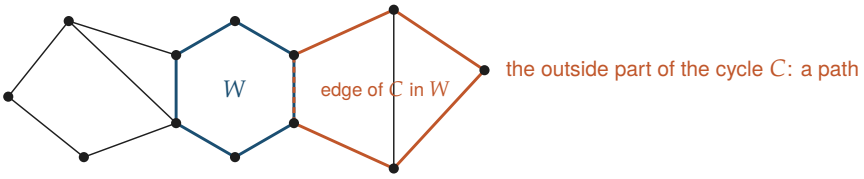


Figure 3. A subgraph W and a cycle C through an edge of W (dashed). The part of C outside W is a forest. An even subgraph whose outside part contains a cycle is not a cycle through W.

In the cases 2, 3 and 4 a cycle consists of paths in the parts of G. By the forest estimate, a part with more than half of the possible paths between two of its ends, by a fixed margin, is a ladder up to a bounded error, and it has few cycles (Section 5). In the case 5 the forest estimate is not sufficient, because we need a bound below $\frac{1}{2}$ and its error term is positive. In this case a second-moment argument gives two disjoint pieces of G with small boundaries and large girth, and each piece contributes a factor of at most $\frac{3}{5}$ to the proportion of forests (Section 8).

1.4 Authorship and verification

KNT is a person. Astra and Fable 5.2 are language models: GPT-6 Astra and Claude Fable 5.2. KNT directed the work at each step. The forest estimate and its proof are from [6]; KNT obtained them in September 2026 with Astra, for a problem of Erdős on pancyclic graphs (see also [8]). In November 2026 the two models studied the number of cycles in parallel, in one shared set of files.

The parts have these origins. Fable 5.2 found the edge recursion, the localisation and layer argument, the chain of nested cycles, the parallel-link lemma and Lemma 6.7. Astra found the path structure of Section 5, the gap theorem, the first analysis of cuts, the count for rings, the argument for bounded girth, and the packing and the two girth scales of Section 8. In its first form, Section 8 gave the bound $\frac{1}{2}$ for the proportion of forests, and the proof of Theorem 1.1 then used an induction on the rank. Astra improved the bound to a constant below $\frac{1}{2}$, and the induction is no longer necessary. Fable 5.2 shortened the arguments for paths and cuts and wrote this text.

Theorem 1.1 has a formal proof in Lean 4. The formal statement is `Erdos1016.ExactMaximum.exactMaximum`: there is a number R such that `maxCyclesr=horakKimr` for every $r \ge R$. Here `maxCyclesr` is the largest number of cycles of a simple graph with cyclomatic number r on a vertex set $\{0, \dots, n-1\}$, a cycle is counted by its edge set, and `horakKimr` is $H(r)$. A reader who trusts the Lean kernel needs to read these definitions only. The table lists the formal statements of the main steps. All of them are proved.

Statement	Declaration
Theorem 3.1	`Erdos1016.ShortProof.fewComponentForestEstimate_logSquared`
Corollary 3.11	`Erdos1016.cycleCountTheorem`; with parallel edges: `EM.numCycles_le_of_large_rank_margin`
Lemma 3.10, as the upper bound uses it	`EMS.CubicReduction.cubicReductionStatement`
Equation (6.1) and the count of Lemma 6.4	`Net.numCycles_of_three_crossing`, `Net.numCycles_of_four_crossing`
Lower bound of Corollary 6.2: for every $r \geq 3$ a graph of rank r with at least $H(r)$ cycles	`EM.lowerBoundStatement`
Case 1 of Section 9 (a loop)	`EM.numCycles_le_horakKim_of_loop`
Theorem 5.4 for $\varepsilon = \frac{1}{1000}$, with Lemma 5.2 and Proposition 5.3; Lemma 5.7	`EM.fewCyclesStatement`, `EM.gadget_third_pair`, `EM.gadget_cross_pairs`, `EM.gadget_opposite_pair`
Case 2: Proposition 6.3 with some constant	`EM.threeCutStatement`
Theorem 5.10	`EMS.gapStatement`
Lemma 6.7; Lemmas 6.5 and 6.6 and the cases of Theorem 6.8, as arithmetic	`Net.two_mul_numPathPairs_le_of_full`, `EMS.FourCutArithmeticProof.fourCutArithmeticStatement`
The last step of Theorem 6.8: four full pairs in both sides give at most $H(r)$ cycles	`EMS.ringCaseStatement`
Case 3: Theorem 6.8 with the rings	`EM.fourCutStatement`
Corollary 5.6, for the pole alone	`EMS.cyclicShoreStatement`
Case 4: Theorem 7.1	`EM.shortCycleStatement`
Case 5 (large girth), with Theorem 8.1	`EMS.LargeGirth.largeGirthStatement`
Upper bound: at most $H(r)$ cycles for large r	`EM.upperBoundStatement`
Theorem 1.1; equation (1.1) for both parities	`EM.exactMaximum`; `EM.maxCycles_formula`

Short names: `EM` is `Erdos1016.ExactMaximum`, `EMS` is `Erdos1016.ExactMaximumSupport`, and `Net` is `Erdos1016.BoundaryTrace.Network`.

The declarations are at commit `3a80aa9` of [9], on the branch `feat/cycle-count-bound`. The files contain no `sorry`: Fable 5.2 wrote `Erdos1016/ExactMaximum/`, and Astra wrote `Erdos1016/ExactMaximumSupport/` and the formal proof of Theorem 3.1. Astra's files contain the formal proofs of Section 5.3, of the upper bound for rings, and of Section 8. The Lean kernel accepts the formal proofs with the axioms `propext`, `Classical.choice` and `Quot.sound` only.

The kernel check covers the formal statements, and no machine has checked the text of this paper. A formal proof can differ from the text in details. For example, the formal proof of Case 5 marks a connected set of eight vertices where the text marks a shortest cycle, the formal proof of Theorem 5.4 calls a thin layer ordinary if at most one of its two vertices is on a pair of parallel edges, and the formal proof of Theorem 7.1 allows $12g + 2$ ports for the poles K_p. So the text can contain errors that the formal proof does not have. The two models have read the text, and no referee has. Scripts in the repository test the statements of Sections 5 and 6 on small graphs.

These tests do not replace a proof.

2 Tools

NOTATION | multigraphs, cuts and regions

- From here on a graph may have loops and parallel edges. We say *multigraph* to stress this, and *simple graph* to exclude it. A loop adds two to the degree of its vertex. A loop is a cycle of length one, and two parallel edges form a cycle of length two.
- A multigraph is *cubic* if every vertex has degree three, and *subcubic* if every vertex has degree at most three. The *girth* is the length of a shortest cycle.
- $c(G)$ is the number of cycles of G, and $c_e(G)$ is the number of cycles through an edge e. If we replace an edge by a path, the rank and the number of cycles do not change. So $\psi(r)$ is also the largest number of cycles of a multigraph of rank r.
- For a set A of vertices, $G[A]$ is the induced subgraph, $G - A$ is $G[V(G) \setminus A]$, and δA is the set of edges with exactly one end in A.
- A *region* is a nonempty set A such that $G[A]$ is connected. A region is *cyclic* if $G[A]$ contains a cycle.
- A *cyclic cut* is a set δX such that $G[X]$ and $G - X$ each contain a cycle; its *size* is $|\delta X|$.
- Throughout, ξ is a uniformly random element of $\mathcal{Z}(G)$.

2.1 Outside parts

Let W be a subgraph of G, with $r = \beta(G)$ and $t = \beta(W)$. For an even subgraph ξ we call $\xi \setminus E(W)$ its *outside part*. The map $\xi \mapsto \xi \setminus E(W)$ is linear on $\mathcal{Z}(G)$ with kernel $\mathcal{Z}(W)$. So there are 2^{r-t} outside parts, each is the outside part of exactly 2^t even subgraphs, and the outside part of a uniform ξ is uniform. Put

$$p(G,W) = \mathbb{P}\big(\xi \setminus E(W) \text{ is a forest}\big), \qquad q(G,P) = \mathbb{P}\big(\xi \cap E(G-P) \text{ is a forest}\big)$$

for a subgraph W and a set P of vertices.

Lemma 2.1 (Edge recursion). Let G be a connected multigraph of rank r, let W be a cycle of G, and let e be an edge of W. Then $G - e$ is connected and has rank $r - 1$, and

$$c(G) = c(G-e) + c_e(G), \qquad c_e(G) \;\le\; p(G,W)\,2^{r-1}.$$

Proof. The edge e lies on a cycle, so $G-e$ is connected with one edge less. A cycle of G avoids e or contains e. This is the first formula.

Let C and C' be cycles through e with the same outside part. Then $C+C'$ is an even subgraph inside W, so it is empty or equal to W. But $C + W$ does not contain e. So $C' = C$. Hence the cycles through e have distinct outside parts. Each of these is a proper subset of a cycle, so it is a forest. There are $p(G,W)\,2^{r-1}$ outside parts that are forests, because $\beta(W) = 1$. ∎

2.2 Cuts and links

For a region A let O_A be the event $\xi \cap \delta A = \emptyset$.

Lemma 2.2. Let $Z \geq 0$ be a random variable with $\mu = \mathbb{E}Z > 0$ and $\mathbb{E}Z^2 \leq 2\mu^2 + K\mu$. Then $\mathbb{P}(Z = 0) \leq \frac{1}{2} + K/(4\mu)$.

Proof. By the Cauchy–Schwarz inequality $\mu^2 \leq \mathbb{E}Z^2 \cdot \mathbb{P}(Z > 0)$, so $\mathbb{P}(Z = 0) \leq 1 - \mu/(2\mu + K) \leq \frac{1}{2} + K/(4\mu)$. ∎

Let A and B be disjoint regions of a connected multigraph G. A *link* between them is an edge with one end in A and the other in B, or a component of $G - (A \cup B)$ that is joined by an edge to A and by an edge to B (Figure 4). Let $\ell(A, B)$ be the number of links.

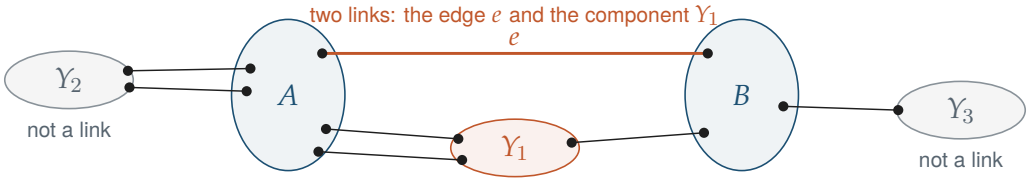


Figure 4. Links between two regions A and B. The edge e and the component Y_1 of $G - (A \cup B)$ are links. The components Y_2 and Y_3 are joined to one region only, so they are not links. Here $\ell(A, B) = 2$.

Lemma 2.3 (Link formula)**.** Let A and B be disjoint regions of a connected multigraph G. Then $\mathbb{P}(O_A) = 2^{\mathrm{comp}(G-A)-|\delta A|} \geq 2^{-|\delta A|}$ and

$$\mathbb{P}(O_A \cap O_B) = 2^{\ell(A,B)-1}\,\mathbb{P}(O_A)\,\mathbb{P}(O_B).$$

Conditionally on O_A, the set $\xi \cap E(G[A])$ is uniform on $\mathcal{Z}(G[A])$. Conditionally on $O_A \cap O_B$, the sets $\xi \cap E(G[A])$ and $\xi \cap E(G[B])$ are independent and uniform on $\mathcal{Z}(G[A])$ and $\mathcal{Z}(G[B])$.

Proof. The even subgraphs that avoid δA are the sums of an even subgraph of $G[A]$ and an even subgraph of $G - A$. So $\mathbb{P}(O_A) = 2^{\beta(G[A])+\beta(G-A)-\beta(G)}$, and a count of edges and vertices shows that the exponent is $\mathrm{comp}(G - A) - |\delta A|$. In the same way

$$\mathbb{P}(O_A \cap O_B) = 2^{\beta(G[A])+\beta(G[B])+\beta(G-A-B)-\beta(G)} = 2^{\mathrm{comp}(G-A-B)+1-|\delta A|-|\delta B|+e(A,B)},$$

where $e(A, B)$ is the number of edges between A and B. Let a, b and k be the numbers of components of $G - A - B$ that are joined to A only, to B only, and to both. Then $\mathrm{comp}(G - A - B) = a + b + k$, $\mathrm{comp}(G - A) = a + 1$ and $\mathrm{comp}(G - B) = b + 1$. The formula follows, because $\ell(A, B) = k + e(A, B)$. The conditional laws are read off from the two decompositions as direct sums. ∎

IDEA | the correlation of two cut events

By Lemma 2.3, two cut events are independent when the regions have one link, and their joint probability is twice the product when the regions have two links. The constant $\frac{1}{2}$ of the forest estimate comes from this factor two, through Lemma 2.2.

Pairs with three or more links are rare:

Lemma 2.4 (Exceptional partners). Let A be a region of a connected multigraph G, and let $B_1, \dots, B_k$ be pairwise disjoint regions, disjoint from A, with $\ell(A, B_i) \geq 3$ for all i. Then $k \leq |\delta A|$.

Proof. Contract A to a vertex a and each B_i to a vertex b_i, and delete the loops that arise. This changes neither $d = |\delta A|$ nor the numbers $\ell(A, B_i)$, because the sets B_j with $j \neq i$ are connected and are contracted inside components of $G - (A \cup B_i)$. Let F be a spanning forest of $G - a$, and for every edge $e = au$ at a attach a new leaf τ_e to u. Call the resulting forest F^+ and its d new leaves *terminals*.

Fix i. Every link between a and b_i yields a path of F^+ from b_i to a terminal. For an edge $e = ab_i$ it is the edge $b_i\tau_e$. For a component Y of $G - \{a, b_i\}$ joined to both, choose an edge $e = au$ with $u \in Y$ and take the path of F^+ from b_i to τ_e; after its first vertex it stays in $Y \cup \{\tau_e\}$. Paths that come from different links start with different edges at b_i, and the union of any two of them is a path between two terminals. Let $F^\circ \subseteq F^+$ be the union of all paths between terminals. As $\ell(a, b_i) \geq 3$, the vertex b_i has degree at least 3 in F°. All leaves of F° are terminals, and a forest with at most d leaves has at most $d - 2$ vertices of degree at least 3. Hence $k \leq d - 2$ if $k \geq 1$. ∎

Lemma 2.5 (Many small cuts). Let $A_1, \dots, A_k$ be pairwise disjoint cyclic regions of a connected multigraph G with $|\delta A_i| \leq d$ for all i. Then

$$\mathbb{P}\big(\xi \cap E(G[A_i]) \text{ is a forest for every } i\big) \;\leq\; \frac{1}{2} + \frac{(d+1)\,2^{d-1}}{k}.$$

Proof. Let E_i be the event that O_{A_i} holds and $\xi \cap E(G[A_i]) \neq \emptyset$. On E_i the set $\xi \cap E(G[A_i])$ is a nonempty even subgraph, hence not a forest. So it suffices to bound $\mathbb{P}(Z = 0)$ for $Z = \sum_i \mathbf{1}_{E_i}$. By Lemma 2.3, $p_i := \mathbb{P}(E_i) = \mathbb{P}(O_{A_i})\big(1 - 2^{-\beta(G[A_i])}\big) \geq 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(A_i, A_j)-1} p_i p_j$ for $i \neq j$. If $\ell(A_i, A_j) \leq 2$, this is at most $2p_ip_j$. Otherwise we use $\mathbb{P}(E_i \cap E_j) \leq p_i$; by Lemma 2.4 this happens for at most d indices j, for each i. Hence $\mathbb{E}Z^2 \leq \mu + 2\mu^2 + d\mu$ with $\mu = \mathbb{E}Z \geq k\,2^{-d-1}$, and Lemma 2.2 gives the claim. ∎

2.3 Localisation

By the next lemma, the part of ξ inside a set X has the same distribution as the part of a random even subgraph of a smaller graph outside one cycle.

Lemma 2.6 (Localisation). Let G be a connected multigraph, and let X be a nonempty proper set of vertices such that $G - X$ is connected. Put $d = |\delta X|$. Form G_X from $G[X]$ as follows. For each edge of δX add a new vertex and join it to the end of that edge in X. Join the new vertices by a cycle W_X through all of them; if $d < 3$, first add $3 - d$ further new vertices on that cycle. Then

(a) G_X is connected, and if every vertex of X has degree three in G, then $\beta(G_X) = (|X| + d)/2 + 1$;

(b) the set $\xi \cap E(G[X])$ has the same distribution for ξ uniform on $\mathcal{Z}(G)$ and for ξ uniform on $\mathcal{Z}(G_X)$;

(c) an even subgraph ξ of G_X has a forest outside $E(W_X)$ if and only if $\xi \cap E(G[X])$ is a forest.

Proof. (a) Every component of $G[X]$ has an edge to $G - X$, so the cycle W_X joins all components. Without the extra vertices, G_X has d more vertices and $2d$ more edges than $G[X]$, so $\beta(G_X) = |E(G[X])| + 2d - |X| - d + 1$. Extra vertices on the cycle add as many edges as vertices. If all degrees in X are three, then $3|X| = 2|E(G[X])| + d$.

(b) Contract the connected set $V(G) \setminus X$ of G to one vertex, and delete the loops that arise. Call the result G/X. An even subgraph of G restricts to an even subgraph of G/X. Every even subgraph of G/X arises in this way: the missing edges inside the contracted set can be chosen in a spanning tree of that set. So the restriction is a surjective linear map $\mathcal{Z}(G) \to \mathcal{Z}(G/X)$, and it sends the uniform distribution to the uniform distribution. Now contract W_X in G_X. The result is the same multigraph G/X, with the new edges in the place of the edges of δX.

(c) The edges of G_X outside W_X are the edges of $G[X]$ and the d new edges at X. In $G_X - E(W_X)$ each of these new edges has an end of degree one, so it lies on no cycle. ∎

3 The forest estimate

Theorem 3.1 (Forest estimate, KNT [6]). There is r_0 such that the following holds for all $r \geq r_0$. Let G be a connected simple graph of rank r. Let $W \subseteq G$ be a nonempty union of cycles with at most $r/\log^2 r$ edges and at most $(\log r)/100$ components. Then

$$p(G, W) \leq \frac{1}{2} + \frac{20}{\log^{(3)} r}. \tag{3.1}$$

This paper uses the theorem for a cycle W only. The form with several components has the same proof, and [8] needs it. The proof is that of [6].

IDEA | the proof in four steps

1. We pass to a multigraph Γ with all degrees at most 3, in which W has become a small set P of vertices (Lemma 3.2).
2. If $\Gamma - P$ has many disjoint short cycles, the estimate follows from Lemma 2.5. So we may delete a small maximal family of them, and a graph J of large girth remains.
3. The graph J has many cycles with between $6 \log r$ and $\log r \sqrt{\log \log r}$ edges. Their total weight $\sum_C 2^{-|C|}$ is at least $(\frac{1}{8} - o(1)) \ln \log \log r$ (Lemma 3.5).
4. For such a cycle C, the event that C is a component of ξ has probability at least $2^{-|C|}$, and then the part of ξ outside P is not a forest. Most pairs of these events are correlated by a factor of at most two (Lemma 2.3).

With the factor two of step 4, the second moment of the number Z of these events is at most $2\mu^2 + K\mu$, and Lemma 2.2 then gives $\mathbb{P}(Z = 0) \le \frac{1}{2} + K/(4\mu)$.

We state the lemmas for the steps 3 and 4 with free parameters, because Section 8 uses them again.

3.1 Reduction to a subcubic multigraph

Lemma 3.2. Let G and W be as in Theorem 3.1, with $m = |E(W)|$ and $c = \mathrm{comp}(W)$. There are a connected multigraph Γ with $\beta(\Gamma) = r$ and all degrees at most 3, and a set $P \subseteq V(\Gamma)$, such that

(i) every vertex outside P has degree 3, $|P| \le 2m$, and $\Gamma[P]$ has at most c components;

(ii) $p(G, W) \le q(\Gamma, P)$.

In particular $2r - 2 \le |V(\Gamma)| \le 2r + 2m$.

Proof. **Splitting.** Replace every vertex v of degree $d \ge 4$ by a path $v^1 v^2 \cdots v^{d-2}$. Distribute the d edges at v along this path in some order: two to v^1, one to each of $v^2, \dots, v^{d-3}$, and two to v^{d-2}. If $v \in V(W)$, let the edges of W at v come first in this order, so that they end in an initial segment $v^1, \dots, v^i$ of the path with $i < \deg_W(v)$. Let G' be the new graph; its degrees are at most 3. Let P consist of these initial segments and of the vertices of W that were not split. Then $|P| \le \sum_v \deg_W(v) = 2m$, every edge of W joins two vertices of P, and $G'[P]$ has at most c components.

Contracting the new path edges maps $\mathcal{Z}(G')$ bijectively onto $\mathcal{Z}(G)$, so we may take $\xi_G = \xi_{G'} \cap E(G)$. Every edge of $G' - P$ is a new path edge or an edge of $E(G) \setminus E(W)$. If $\xi_G \setminus E(W)$ is a forest, then so is its union with the set of all new path edges, because contracting these edges, which form a forest, gives back $\xi_G \setminus E(W)$. Hence $p(G, W) \le q(G', P)$.

Cleaning. As long as possible, delete a vertex of degree one outside P, or replace a vertex of degree two outside P and its two edges by a single edge. An even subgraph avoids a pendant edge and contains both or none of the two edges at a vertex of degree two. So

these operations induce bijections of the cycle spaces. Let Γ be the final multigraph. Each edge of $\Gamma - P$ is a path of G' all of whose vertices lie outside P. So if ξ_Γ contains a cycle of $\Gamma - P$, then the corresponding $\xi_{G'}$ contains a cycle of $G' - P$. This proves (ii).

Edges with both ends in P are never removed. So the vertices of P keep degree at least two, and the number of components of the graph induced on P does not increase. The vertices outside P now have degree exactly 3. Finally $3|V(\Gamma)| - |P| \le 2|E(\Gamma)| \le 3|V(\Gamma)|$ and $|E(\Gamma)| = |V(\Gamma)| + r - 1$ give the bounds on $|V(\Gamma)|$. ■

3.2 Cycles of moderate length in a graph of large girth

In this subsection Γ is a connected multigraph with all degrees at most 3, and U is a set of its vertices such that every vertex outside U has degree 3. We suppose that $J = \Gamma - U$ is a simple graph of girth greater than D, where $D \ge 3$ is an integer, and we put $s = \lfloor (D-1)/2 \rfloor$. Let $L \ge 1$ be an integer, and let $\mathcal{C}$ be the set of the cycles of J of length at most L. Let v be the number of vertices of J, and let

$$d_J = \sum_{u \in V(J)} \bigl(3 - \deg_J u\bigr)$$

be its *deficit*. This is the number of edges between $V(J)$ and U, so $d_J \le 3|U|$.

A walk $u_0 u_1 \cdots u_\ell$ is *non-backtracking* if $u_{i+1} \ne u_{i-1}$ for $0 < i < \ell$. Let $N_\ell(u, w)$ be the number of non-backtracking walks of length ℓ from u to w in J.

Lemma 3.3. If $\ell > s$, then $N_\ell(u, w) \le 3 \cdot 2^{\ell - s - 1}$ for all u, w. Consequently, for every vertex w of J,

$$\sum_{C \in \mathcal{C},\ C \ni w} 2^{-|C|} \ \le\ L\, 2^{-s}. \tag{3.2}$$

Proof. There are at most $3 \cdot 2^{\ell - s - 1}$ ways to choose the first $\ell - s$ steps. The last s steps form a non-backtracking walk of length s between two given vertices. Such a walk is a path, because s is less than the girth. There is at most one such path, because two of them would give a cycle of length at most $2s < D$. For the second statement, a cycle of length ℓ through w can be traversed from w in two directions. So there are at most $\frac{1}{2} N_\ell(w, w)$ of them, and $\ell > D > s$. Hence the cycles of each length $\ell \le L$ contribute at most $\frac{3}{4}\, 2^{-s}$. ■

Let B be the non-backtracking matrix of J. Its rows and columns are indexed by the directed edges $(i \to j)$ of J, two for each edge. The entry in row $(i \to j)$ and column $(j' \to k)$ is 1 if $j' = j$ and $k \ne i$, and it is 0 otherwise.

Lemma 3.4. Suppose that $d_J < v$. Then, for every even $\ell \ge 2$,

$$\operatorname{tr} B^\ell \ \ge\ \bigl(2 - d_J/v\bigr)^\ell - 3v\, 2^{\ell/2}.$$

Proof. Let A be the adjacency matrix of J and Δ the diagonal matrix of its degrees, and put $M(\lambda) = \lambda^2 I - \lambda A + \Delta - I$.

(1) *If $M(\lambda)g = 0$ for a real $\lambda > 1$ and a vector $g \neq 0$, then λ is an eigenvalue of B.* A direct computation shows that $f(i \to j) = \lambda g(j) - g(i)$ satisfies $Bf = \lambda f$. And $f = 0$ would give $g(i) = \lambda g(j) = \lambda^2 g(i)$ along every edge, hence $g = 0$; at an isolated vertex, the equation $M(\lambda)g = 0$ is $(\lambda^2 - 1)g = 0$.

(2) *If $Bf = \lambda f$ with $f \neq 0$ and $\lambda^2 \neq 1$, then $M(\lambda)g = 0$ for $g(j) = \sum_{k\sim j} f(j \to k)$, and $g \neq 0$.* The eigenvalue equation at $(i \to j)$ and at $(j \to i)$ reads $\lambda f(i \to j) = g(j) - f(j \to i)$ and $\lambda f(j \to i) = g(i) - f(i \to j)$. Eliminating $f(j \to i)$ gives $(\lambda^2 - 1)f(i \to j) = \lambda g(j) - g(i)$. This shows that $g \neq 0$. Summing it over the neighbours j of i gives $M(\lambda)g = 0$.

(3) *B has a real eigenvalue $\lambda_1 \geq 2 - d_J/v$.* The matrix $M(2) = 2(\Delta - A) + (3I - \Delta)$ is positive semidefinite, while $\mathbf{1}^\top M(\lambda)\mathbf{1} = (\lambda - 1)(v\lambda - 2v + d_J)$ vanishes at $\lambda^* = 2 - d_J/v > 1$. The smallest eigenvalue of the symmetric matrix $M(\lambda)$ is continuous in λ, at most 0 at λ^* and at least 0 at 2. So $M(\lambda_1)$ is singular for some $\lambda_1 \in [\lambda^*, 2]$, and (1) applies.

(4) *Every non-real eigenvalue λ of B satisfies $|\lambda| \leq \sqrt{2}$.* With g as in (2), $g^* M(\lambda)g = 0$ is a quadratic equation $a\lambda^2 - b\lambda + c' = 0$ with real coefficients $a = g^*g > 0$, $b = g^*Ag$ and $c' = g^*(\Delta - I)g \leq 2a$. Its roots are λ and $\bar\lambda$, so $|\lambda|^2 = c'/a \leq 2$.

Now $\operatorname{tr} B^\ell$ is the sum of the ℓth powers of the $2|E(J)| \leq 3v$ eigenvalues of B. For even ℓ the real eigenvalues contribute nonnegative terms, one of which is at least $(2 - d_J/v)^\ell$, and the non-real ones contribute at least $-3v\, 2^{\ell/2}$ in total. ∎

Facts (2) and (4) are part of the Ihara–Bass theory [3, 7]. The test vector $\mathbf{1}$ in (3) is from [6]. With this test vector the argument needs no assumption of regularity or expansion.

Lemma 3.5 (Cycle weight). Suppose that $d_J < v$, and let $\ell_0 \leq L$ be a positive integer. The total weight $\mu := \sum_{C\in\mathcal{C}} 2^{-|C|}$ satisfies

$$\left(1 + \tfrac{3}{2} L^2 2^{-s}\right)\mu \;\geq\; \left(1 - \frac{L\, d_J}{2v} - 3v\, 2^{-\ell_0/2}\right) \sum_{\ell} \frac{1}{2\ell},$$

where the sum is over the even ℓ with $\ell_0 \leq \ell \leq L$.

Proof. $\operatorname{tr} B^\ell$ is the number of closed walks $u_0 u_1 \cdots u_\ell = u_0$ in J with $u_{i+1} \neq u_{i-1}$ for all i modulo ℓ. A cycle of length ℓ gives 2ℓ of them; let n_ℓ be the number of the others. Then

$$T := \sum_{\ell \leq L} \frac{\operatorname{tr} B^\ell}{2\ell\, 2^\ell} = \mu + \sum_{\ell \leq L} \frac{n_\ell}{2\ell\, 2^\ell}.$$

Consider a walk counted by n_ℓ, and let j be the least index with $u_j \in \{u_0, \ldots, u_{j-1}\}$, say $u_j = u_i$. Then $j < \ell$, the walk $C = u_i u_{i+1} \cdots u_j$ is a cycle, of length q say, and $u_j u_{j+1} \cdots u_\ell u_1 \cdots u_i$ is a non-backtracking closed walk of length $\ell - q > 0$ from u_i to itself. It contains a cycle, so $\ell - q > D > s$. The original walk is determined by C, by the vertex u_i and a direction on C

($2q$ choices), by the complementary walk (at most $3 \cdot 2^{\ell-q-s-1}$ choices, by Lemma 3.3), and by the position of u_0 on the latter (at most ℓ choices). Hence

$$\sum_{\ell \le L} \frac{n_\ell}{2\ell\, 2^\ell} \;\le\; \sum_{\ell \le L} \sum_{C \in \mathcal{C}} \tfrac{3}{2}\, |C|\, 2^{-|C|-s} \;\le\; \tfrac{3}{2}\, L^2\, 2^{-s}\, \mu,$$

and $T \le (1 + \frac{3}{2}L^2 2^{-s})\mu$. On the other hand, all terms of T are nonnegative. For even ℓ with $\ell_0 \le \ell \le L$, Lemma 3.4 gives

$$2^{-\ell} \operatorname{tr} B^\ell \;\ge\; \Big(1 - \frac{d_J}{2v}\Big)^\ell - 3v\, 2^{-\ell/2} \;\ge\; 1 - \frac{L\, d_J}{2v} - 3v\, 2^{-\ell_0/2}. \qquad \blacksquare$$

The second moment. Let ξ be uniform on $\mathcal{Z}(\Gamma)$. For $C \in \mathcal{C}$ let I_C be the event that C is a component of ξ. As the vertices of C have degree 3 in Γ, this is the event $E(C) \subseteq \xi$. It is also the event that $O_{V(C)}$ holds and $\xi \cap E(\Gamma[V(C)]) = E(C)$. Put

$$a_C = 2^{-|C|}, \qquad p_C = \mathbb{P}(I_C), \qquad X_C = \frac{a_C}{p_C}\, \mathbf{1}_{I_C}, \qquad Z = \sum_{C \in \mathcal{C}} X_C.$$

Because of the factor a_C/p_C, the mean of X_C is the combinatorial weight a_C for every value of p_C. The event I_C is given by $|C|$ linear conditions on $\xi \in \mathcal{Z}(\Gamma)$, which are satisfied by $\xi = C$. So $p_C \ge a_C$. Hence $0 \le X_C \le 1$, and $\mathbb{E}Z = \mu$. If $Z \ne 0$, then ξ contains a cycle of J.

Let $C \ne C'$ be in $\mathcal{C}$. If they share a vertex, then $I_C \cap I_{C'} = \emptyset$. If they are disjoint, Lemma 2.3, applied to the regions $V(C)$ and $V(C')$ of Γ, gives $\mathbb{P}(I_C \cap I_{C'}) = 2^{\ell(C,C')-1} p_C p_{C'}$ and therefore

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'}\, 2^{\ell(C,C')-1}, \tag{3.3}$$

where $\ell(C, C')$ is the number of links between $V(C)$ and $V(C')$ in Γ.

Lemma 3.6 (Second moment). Suppose that $\ell(C, C') \le b$ for all disjoint $C, C' \in \mathcal{C}$. Let W_2 be the sum of $a_C a_{C'}$ over the ordered pairs of disjoint cycles of $\mathcal{C}$ with exactly two links. Then $W_2 \le \mu^2$ and

$$\mathbb{E}Z^2 \;\le\; \mu^2 + W_2 + \big(1 + 2^b L^3 2^{-s}\big)\mu.$$

Proof. Call a disjoint pair $C, C' \in \mathcal{C}$ *exceptional* if $\ell(C, C') \ge 3$. Fix C, and let $\mathcal{E}$ be a maximal family of pairwise disjoint exceptional partners of C. By Lemma 2.4, $|\mathcal{E}| \le |\delta V(C)| \le L$. So the union of their vertex sets has at most L^2 vertices, and every exceptional partner of C meets this union. By (3.2), the sum of $a_{C'}$ over the exceptional partners C' of C is at most $L^3 2^{-s}$. With (3.3),

$$\sum_{C' \text{ exceptional}} \mathbb{E}(X_C X_{C'}) \;\le\; a_C\, 2^b L^3\, 2^{-s}.$$

By (3.3), a disjoint pair with at most one link has $\mathbb{E}(X_C X_{C'}) \le a_C a_{C'}$, and a pair with two links has $\mathbb{E}(X_C X_{C'}) = 2a_C a_{C'}$. Intersecting pairs contribute nothing, and $\mathbb{E}X_C^2 \le \mathbb{E}X_C = a_C$. Sum over all ordered pairs. $\blacksquare$

Lemma 3.7 (Links through trees). Let $8 \le D \le L$, and let t be an integer with $1 \le t \le D/4$. Let $C, C' \in \mathcal{C}$ be disjoint, and put $U' = V(C) \cup V(C')$. Consider the links between $V(C)$ and $V(C')$ of two kinds: the edges; and the components Y of $\Gamma - U'$ that avoid U, induce a tree, and have at most t edges to U'. There are at most $289(L/D)^2$ such links.

Proof. Let Y be a link of the second kind, with $d_Y \le t$ edges to U'. All its vertices have degree 3, so $3|Y| = 2(|Y| - 1) + d_Y$, that is, $|Y| = d_Y - 2$. So there is a path from C to C' through Y with at most t edges. For a link that is an edge, take this edge as the path. These paths lie in J. They are pairwise disjoint, because every vertex of U' lies on exactly one edge outside its own cycle. Divide each of C and C' into arcs of at most $\lfloor D/8 \rfloor$ consecutive vertices. At most $17L/D$ arcs are necessary for each cycle, because $\lfloor D/8 \rfloor \ge D/16$ and $L \ge D$. If two of the paths had their ends in the same arc of C and in the same arc of C', they would form, together with parts of these two arcs, a cycle of J of length at most $2t + 2\lfloor D/8 \rfloor < D$. So there are at most $(17L/D)^2$ paths. ∎

3.3 Proof of Theorem 3.1

Let Γ and P be as in Lemma 3.2, let ξ be uniform on $\mathcal{Z}(\Gamma)$, and let r be large. All $o(\cdot)$ and $O(\cdot)$ refer to $r \to \infty$ and are uniform in G and W. Put

$$x = \log r, \qquad y = \log x, \qquad z = \log y,$$
$$D = \lfloor x/10 \rfloor, \qquad L = \lfloor x\sqrt{y} \rfloor, \qquad s = \lfloor (D-1)/2 \rfloor, \qquad \varepsilon = 20/z.$$

Recall that $|P| \le 2r/x^2$ and $c \le x/100$. Let $\mathcal{F}$ be the event that $\xi \cap E(\Gamma - P)$ is a forest. We assume that

$$\mathbb{P}(\mathcal{F}) > \tfrac{1}{2} + \varepsilon \tag{3.4}$$

and derive a contradiction. By Lemma 3.2(ii) this proves Theorem 3.1.

Removing the short cycles. Let $\mathcal{Q}$ be a maximal family of pairwise disjoint cycles of $\Gamma - P$ of length at most D, loops and pairs of parallel edges included, and let S be the union of their vertex sets. For $Q \in \mathcal{Q}$ the set $V(Q)$ is a cyclic region of Γ with $|\delta V(Q)| \le |Q| \le D$, because its vertices have degree 3. On $\mathcal{F}$ the set $\xi \cap E(\Gamma[V(Q)])$ is a forest for every Q. So Lemma 2.5 and (3.4) give $|\mathcal{Q}| < (D+1)2^{D-1}/\varepsilon \le \sqrt{r}$. Hence $|S| \le D\sqrt{r}$.

Put $U = P \cup S$ and $J = \Gamma - U$. By the maximality of $\mathcal{Q}$, J is a simple graph of girth greater than D. So we are in the setting of Section 3.2. The number v of vertices of J satisfies $v = (2 + o(1))r$, and

$$d_J \;\le\; 3\big(|P| + |S|\big) \;\le\; \frac{7r}{x^2}. \tag{3.5}$$

The weight of the cycles. We claim that

$$\mu \;\ge\; \big(\tfrac{1}{8} - o(1)\big)\ln y. \tag{3.6}$$

This is Lemma 3.5 with $\ell_0 = \lceil 6x \rceil$. Indeed $2^{-s} \le 8\, r^{-1/20}$ and $L^2 \le x^2 y$, so $L^2 2^{-s} = o(1)$. By (3.5), $L\, d_J/(2v) = O(\sqrt{y}/x)$. Also $3v\, 2^{-\ell_0/2} \le 3v\, r^{-3} = o(1)$. The sum of $1/(2\ell)$ over the even ℓ in $[6x, L]$ is $(\frac{1}{4} - o(1))\ln(L/6x) = (\frac{1}{8} - o(1))\ln y$.

On $\mathcal{F}$ no cycle of $J \subseteq \Gamma - P$ is contained in ξ, so $Z = 0$. By (3.4),

$$\mathbb{P}(Z = 0) > \tfrac{1}{2} + \varepsilon. \tag{3.7}$$

Lemma 3.8 (Few links). $\ell(C, C') \le c + o(x)$ for all disjoint $C, C' \in \mathcal{C}$.

Proof. Let $U' = V(C) \cup V(C')$. Every vertex of U' lies on exactly one edge outside its own cycle, so at most $2L$ edges leave U'. At most c components of $\Gamma - U'$ meet P, because $P \cap U' = \emptyset$ and so each component of $\Gamma[P]$ lies in one of them. Let Y run over the other components and let d_Y be the number of edges between Y and U'; then $\sum_Y d_Y \le 2L$. Put $t_0 = \lfloor y/2 \rfloor$.

Large cuts. At most $2L/t_0$ components have $d_Y > t_0$.

Small cuts, with a cycle. The components Y with $d_Y \le t_0$ that contain a cycle are pairwise disjoint cyclic regions inside $\Gamma - P$ with $|\delta Y| \le t_0$. On $\mathcal{F}$ each $\xi \cap E(\Gamma[Y])$ is a forest, so by Lemma 2.5 and (3.4) there are fewer than $(t_0 + 1)2^{t_0 - 1}/\varepsilon$ of them.

Small cuts, trees. Let Y induce a tree, with $d_Y \le t_0$. No $Q \in \mathcal{Q}$ meets Y, since Q is a connected subgraph of $\Gamma - U'$ and would lie in Y. Hence Y avoids U. By Lemma 3.7 with $t = t_0$, at most $289(L/D)^2$ links are such components or edges.

Altogether

$$\ell(C, C') \;\le\; c + \frac{2L}{t_0} + \frac{(t_0 + 1)2^{t_0 - 1}}{\varepsilon} + 289\Big(\frac{L}{D}\Big)^2 = c + O\Big(\frac{x}{\sqrt{y}}\Big) + O\big(yz\sqrt{x}\big) + O(y) = c + o(x). \qquad \blacksquare$$

The contradiction. By Lemma 3.8 we may use Lemma 3.6 with $b = c + o(x)$. As $c \le x/100$,

$$2^b L^3 2^{-s} = 2^{c - x/20 + o(x)} = o(1).$$

So $\mathbb{E}Z^2 \le 2\mu^2 + (1 + o(1))\mu$. By Lemma 2.2 and (3.6),

$$\mathbb{P}(Z = 0) \;\le\; \frac{1}{2} + \frac{1 + o(1)}{4\mu} \;\le\; \frac{1}{2} + \frac{2 + o(1)}{\ln y} \;<\; \frac{1}{2} + \frac{20}{z}$$

for large r, since $\ln y = z \ln 2$. This contradicts (3.7), and Theorem 3.1 is proved.

3.4 The forest input and the leading term

Later sections use the theorem for one cycle in a multigraph, with a fixed error.

Proposition 3.9 (Forest input). For every $\delta \in (0, \frac{1}{2})$ there is an integer $R_\delta \ge 2^{100}$ with the following property. Let H be a connected multigraph of rank $\rho \ge R_\delta$, and let W be a cycle of H with at least three and at most $\rho/\log^2 \rho$ edges. Then $p(H, W) \le \frac{1}{2} + \delta$.

Proof. Replace every edge of H outside W by a path with three edges. The new graph H' is simple and connected, it contains W, and it has rank ρ. The even subgraphs of H' cor-

respond to those of H, and one has a forest outside W if and only if the other has. Apply Theorem 3.1 to H' and W. Choose R_δ so large that $\rho \geq R_\delta$ is in the range of the theorem, $\log \rho \geq 100$, and $20/\log^{(3)} \rho \leq \delta$. ∎

Lemma 3.10 (Cubic multigraphs). For every $r \geq 2$ there is a connected cubic multigraph of rank r with exactly $\psi(r)$ cycles.

Proof. The numbers of cycles of the graphs of rank r are integers below 2^r, so some graph G_0 attains $\psi(r)$. If G_0 is not connected, identify a vertex of one component with a vertex of another. This changes neither the rank nor the set of cycles. Now delete a vertex of degree one, or replace a vertex of degree two and its two edges by one edge, as long as this is possible. These steps keep the rank and the number of cycles. The result is a connected multigraph with all degrees at least three, because $r \geq 2$.

Let v be a vertex of degree $d \geq 4$. Replace v by a path $v^1 v^2 \cdots v^{d-2}$, and distribute the d edge ends at v along this path: two at v^1, one at each of $v^2, \ldots, v^{d-3}$, and two at v^{d-2}. The new multigraph has $d-3$ more vertices and $d-3$ more edges, so its rank is r. A cycle through v uses two edge ends at v. Together with the part of the path between these two ends it is a cycle of the new multigraph. Different cycles give different cycles, and the cycles that avoid v remain. So the number of cycles does not decrease, and it cannot exceed $\psi(r)$. Repeat this for every vertex of degree at least four. ∎

Corollary 3.11 (Leading term). There is r_1 such that every graph of rank $r \geq r_1$ has at most $\left(\frac{1}{2} + 21/\log^{(3)} r\right)2^r$ cycles. Consequently $\psi(r) \sim 2^{r-1}$.

Proof. Let r be large, and let G be a connected cubic multigraph of rank r with $\psi(r)$ cycles (Lemma 3.10). Let W be a shortest cycle of G, of length g. A breadth-first search from a vertex meets no vertex twice up to distance $\lfloor (g-1)/2 \rfloor$, so G has at least $2^{(g-2)/2}$ vertices. As G has $2r-2$ vertices,

$$g \;\leq\; 2\log(2r) + 2. \tag{3.8}$$

If $g < 3$, subdivide an edge of W twice; this changes neither $c(G)$ nor $p(G, W)$. Now W has between 3 and $r/\log^2 r$ edges. Subdivide the edges outside W as in the proof of Proposition 3.9. Then Theorem 3.1 gives $p(G, W) \leq \frac{1}{2} + \delta_r$ with $\delta_r = 20/\log^{(3)} r$. Let e be an edge of W. By Lemma 2.1,

$$\psi(r) = c(G) \;\leq\; c(G-e) + \left(\tfrac{1}{2} + \delta_r\right)2^{r-1} \;\leq\; \psi(r-1) + \left(\tfrac{1}{2} + \delta_r\right)2^{r-1}.$$

Put $y_r = \psi(r)2^{-r} - \frac{1}{2}$. Then $y_r \leq \frac{1}{2}y_{r-1} + \frac{1}{2}\delta_r$ for all large r. The numbers δ_r decrease, and $y_r \leq \frac{1}{2}$. So $y_r \leq 2^{-j}y_{r-j} + \delta_{r-j}$ for $j \leq r/2$. With $j = \lfloor r/2 \rfloor$ this is at most $21/\log^{(3)} r$ when r is large. Finally $\psi(r) > 2^{r-1}$ for $r \geq 3$ [4]; the rings of Section 6.1 also show this. ∎

4 Layers

The later sections use Theorem 3.1 through Corollary 3.11 and through one lemma.

IDEA | the content of the lemma

If more than half of the outside parts are forests, by a fixed margin, then every distance layer around the marked set is small. In the other case a window of consecutive layers is a large set with a small boundary, and the forest estimate applies inside this set. We use the lemma for paths in Section 5 and for large girth in Section 8.

Let G be a multigraph and P a set of vertices. A *level function* is a map $\lambda\colon V(G)\to\mathbb{Z}$ such that $|\lambda(u)-\lambda(w)|\le 1$ for every edge uw that has an end outside P. The distance from a fixed vertex is a level function. For $\delta>0$ let R_δ be as in Proposition 3.9, and for an integer $k\ge 1$ put

$$\Theta_\delta(k)=\max\bigl\{2R_\delta,\ 2^{100}(k+1)^2\bigr\}.$$

Lemma 4.1 (Layers). Let G be a connected subcubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and every vertex outside P has degree three. Let λ be a level function such that at most k vertices of P have any given level. Let $\delta\in(0,\frac12)$, and suppose that $q(G,P)>\frac12+\delta$. Then, for every j, fewer than $\Theta_\delta(k)$ vertices outside P have level j.

Proof. Put $\Phi(m)=m/\log^2 m$ and $\Theta=\Theta_\delta(k)$.

Step 1: no large set outside P has a small boundary. We claim that no set $X\subseteq V(G)\setminus P$ has $|X|=m\ge\Theta$ and $|\delta X|\le\Phi(m)/4$. Suppose that X is such a set. Let C_0 be the component of $G-X$ that contains P, and put $X^+=V(G)\setminus V(C_0)$. Then $X\subseteq X^+$, the set X^+ avoids P, and $G-X^+=C_0$ is connected. Different components of $G-X$ have no edge between them, so $\delta X^+\subseteq\delta X$. Every vertex of X^+ has degree three. Let $d^+=|\delta X^+|\ge 1$ and apply Lemma 2.6 to X^+. The rank of G_{X^+} is

$$\rho=\frac{|X^+|+d^+}{2}+1\ \ge\ \frac{m}{2}\ \ge\ R_\delta.$$

The cycle W_{X^+} has $\max\{d^+,3\}$ edges. The function Φ is increasing on $[8,\infty)$, and $\Phi(m)/4\le\Phi(m/2)$. So $d^+\le\Phi(m)/4\le\Phi(m/2)\le\Phi(\rho)$, and also $3\le\Phi(\rho)$. By Proposition 3.9 and Lemma 2.6(b),(c),

$$\mathbb{P}\bigl(\xi\cap E(G[X^+])\text{ is a forest}\bigr)\ \le\ \frac12+\delta.$$

But $G[X^+]$ is a subgraph of $G-P$, so this probability is at least $q(G,P)>\frac12+\delta$. This contradiction proves the claim.

Step 2: a window of levels. Let m_j be the number of vertices outside P of level j. For $i\le i'$ let $X(i,i')$ be the set of vertices outside P with a level in $[i,i']$. An edge that leaves $X(i,i')$ ends outside P at level $i-1$ or $i'+1$, or it ends in P at a level in $[i-1,i'+1]$. So

$$|\delta X(i,i')|\ \le\ 3m_{i-1}+3m_{i'+1}+3k(i'-i+3). \tag{4.1}$$

Suppose that $m_j \geq \Theta$ for some j. Start with the window $i = i' = j$. As long as a level next to the window has more than $m/(48\log^2 m)$ vertices outside P, where $m = |X(i,i')|$ is the present size, add that level to the window. The graph is finite, so this stops. Let μ be the final size and $\gamma = \log \mu \geq 100$. Each step multiplies the size by more than $1 + 1/(48\gamma^2)$. So the number of steps is at most

$$\frac{\ln \mu}{\ln\bigl(1 + 1/(48\gamma^2)\bigr)} \;\leq\; 96\gamma^2 \ln \mu \;<\; 67\gamma^3.$$

At the end both levels next to the window have at most $\mu/(48\gamma^2)$ vertices outside P. By (4.1),

$$|\delta X(i,i')| \;\leq\; \frac{\mu}{8\gamma^2} + 3k\bigl(67\gamma^3 + 3\bigr) \;\leq\; \frac{\mu}{4\gamma^2} = \frac{\Phi(\mu)}{4}.$$

The second inequality is equivalent to $24k\gamma^2(67\gamma^3+3) \leq 2^\gamma$. This holds because $k \leq 2^{\gamma/2-50}$, by $\mu \geq \Theta \geq 2^{100}(k+1)^2$, and because $24\gamma^2(67\gamma^3+3) \leq 2000\gamma^5 \leq 2^{\gamma/2+50}$ for $\gamma \geq 100$. So the final window contradicts Step 1. ■

5 Poles with many paths

A cut of a cubic graph divides it into two parts, and a cycle that crosses the cut is made of paths in the two parts. So we need to count paths between prescribed ends.

IDEA | the results of this section

First, if more than half of the possible paths between two ends are present, by a fixed margin, then the part looks like a ladder, and it has few cycles (Theorem 5.4). Second, the number of paths between two ends is the largest possible, or it is smaller by a fixed factor (Theorem 5.10).

5.1 Poles

A *pole* is a connected multigraph without loops in which every vertex has degree two or three. Its vertices of degree two are its *ports*. A *k-pole* is a pole with exactly k ports. By a count of edge ends, a k-pole of rank h has $2h + k - 2$ vertices.

For two vertices $s \neq t$ of a multigraph N let $P_{st}(N)$ be the number of paths from s to t. An *$\{s,t\}$-join* is a set of edges in which s and t have odd degree and all other vertices have even degree. If N is connected of rank b, there are exactly 2^b of them: they exist, and the sum of two of them is even. Every path from s to t is an $\{s,t\}$-join, so

$$P_{st}(N) \;\leq\; 2^b. \tag{5.1}$$

Call the pair s,t *full* in N if $P_{st}(N) = 2^{\beta(N)}$.

Lemma 5.1 (Joins). Let N be a pole of rank b, and let s, t be two of its ports. Every $\{s,t\}$-join is the union of a path from s to t and of cycles, and these are pairwise vertex-disjoint. So the pair is full if and only if every cycle of N meets every path from s to t. If some cycle of N meets no path from s to t, then $P_{st}(N) \le 2^{b-1}$.

Proof. In a join, s and t have odd degree at most two, and every other vertex has even degree at most three. So the degrees are one at s and t and zero or two elsewhere. If a cycle Z meets no path from s to t, then $J \mapsto J + Z$ maps the paths injectively to joins that are not paths. ■

Lemma 5.2 (Reduction). Let N be a pole of rank b, and let s, t be two of its ports with $P_{st}(N) > 2^{b-1}$.

(a) Replace each port other than s, t, with its two edges, by one edge. No loop arises. The result N° is a 2-pole with ports s, t. It has the same rank, the same number of paths from s to t, and the same number of cycles as N.

(b) Every bridge of N° separates s from t and is incident with neither. Let $N_1, \dots, N_q$ be the components of N° without its bridges, in the order from s to t. Each N_i is a bridgeless 2-pole of rank $b_i \ge 1$, and

$$b = \sum_i b_i, \qquad P_{st}(N) = \prod_i P(N_i), \qquad c(N) = \sum_i c(N_i),$$

where $P(N_i)$ is the number of paths between the two ports of N_i. In particular $P(N_i)\, 2^{-b_i} \ge P_{st}(N)\, 2^{-b}$ for every i.

Proof. (a) Let v be a port other than s, t. Suppose that both edges at v end in the same vertex u. Then u has degree three, its third edge is a bridge, and no path from s to t meets the cycle through u and v. This contradicts Lemma 5.1. So the two neighbours of v are distinct, and we may replace v by an edge. Paths and cycles correspond. Repeat.

(b) Let e be a bridge of N°, and let Y be a component of $N^\circ - e$, with n vertices, j of them ports. Counting edge ends gives $\beta(Y) = (n - j + 1)/2$. This is a positive integer, so Y contains a cycle. If Y contains neither port or both ports, a cycle on one side of e meets no path from s to t. If e is incident with s, every path from s to t starts with e and avoids the cycle on the side of s. Both cases contradict Lemma 5.1. So every path from s to t passes all bridges, in the same order.

The multigraph N_i is connected and bridgeless. Its vertices have degree three, except the ends of the two bridges at N_i, or s, or t. These two vertices are distinct: a common end of two bridges would have a third edge that is a bridge and does not separate s from t. So N_i is a 2-pole, and it contains a cycle. Ranks add over the N_i, every cycle lies in one N_i, and a path from s to t is a concatenation of paths between the ports of the N_i and of the bridges. The last statement holds because $P(N_j) \le 2^{b_j}$ for all j. ■

5.2 The shape of a pole with many paths

Let N be a bridgeless 2-pole with ports s, t and rank $b \geq 2$. It has $2b$ vertices. It has no cut vertex: a cut vertex of a subcubic multigraph is joined to some component by a single edge. We use the following notation. The *layer* L_j is the set of vertices at distance j from s. Every edge joins vertices of the same layer or of consecutive layers. Let ℓ be the distance from s to t, let $A = w_0 w_1 \cdots w_\ell$ be a shortest path from s to t, and put $J = N - V(A)$. Then $w_j \in L_j$. A vertex of A lies on two edges of A, or on one if it is a port. So it has at most one *further edge*. This edge leads to J, or it is parallel to an edge of A, because A is a shortest path. For $0 < j < \ell$ the layer L_j separates s from t, so $|L_j| \geq 2$. Call L_j *thin* if $0 < j < \ell$ and $|L_j| = 2$. A thin layer consists of w_j and one vertex of J. Call a thin layer *ordinary* if its vertex of J is on no pair of parallel edges (Figure 5).

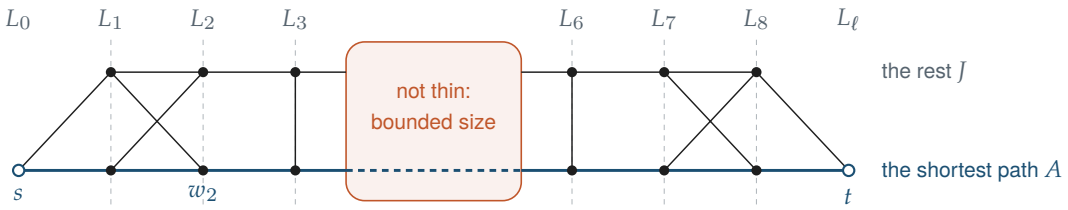


Figure 5. A bridgeless 2-pole with many paths from s to t (schematic). Almost every layer is thin: it has one vertex of the shortest path A and one vertex of J (Proposition 5.3).

IDEA | outline of the proof of Proposition 5.3

Add the shortest path A to a path from s to t. The result is an even subgraph whose edges away from A form a forest. So more than half of the outside parts are forests, and every layer is small by Lemma 4.1. By Lemma 2.5 there are few short cycles away from A. By a count of vertices, almost every layer then has exactly two vertices.

Proposition 5.3 (Thin layers). For every $\varepsilon \in (0, \frac{1}{2}]$ there is a number $C = C_\varepsilon \geq 1$ with the following property. Let N be a bridgeless 2-pole of rank $b \geq 2$ with ports s, t and $P_{st}(N) \geq (\frac{1}{2} + \varepsilon)2^b$. Then, in the notation above,

(i) $\beta(J) \leq C$ and $b - C \leq \ell \leq b$;

(ii) at most $2C$ of the layers L_j with $0 < j < \ell$ are not thin, and $L_j = \emptyset$ for $j > b + C$;

(iii) at most $10C + 2$ vertices of N lie outside the ordinary layers.

Proof. Step 1: the layers are small. Put $P = V(A)$. Then $N[P]$ is connected, and every vertex outside P has degree three. For a path Q from s to t, the set $Q + A$ is an even subgraph of N. Its edges in $N - P$ are edges of the path Q, so they form a forest. Different paths give different even subgraphs, and N has 2^b even subgraphs. Hence

$$q(N, P) \geq \frac{P_{st}(N)}{2^b} \geq \frac{1}{2} + \varepsilon. \tag{5.2}$$

Let $\lambda(u)$ be the distance from s to u in N. This is a level function, and P has at most one vertex of each level. By Lemma 4.1 with $\delta = \varepsilon/2$, every layer L_j has fewer than $\Theta := \Theta_{\varepsilon/2}(1)$ vertices of J.

Step 2: a short cycle. Put $U_j = L_j \setminus V(A)$, and for an interval I of indices let J_I be the subgraph of J induced by the sets U_j with $j \in I$. Suppose that J_I contains a cycle. Choose a cycle Z of J_I such that the interval $[a, a']$ of the layers that Z meets is as short as possible, and put $\sigma = a' - a$ and $R = U_a \cup \cdots \cup U_{a'}$. We claim that $|R| \le 20\Theta$.

By the choice of Z, every cycle of $J[R]$ meets U_a. So $J[R] - U_a$ is a forest. Deleting a vertex of degree at most three lowers the rank by at most two, so $\beta(J[R]) \le 2|U_a| < 2\Theta$. The vertices of R have degree three in N. Counting edge ends gives $|R| = 2\beta(J[R]) + |\delta R| - 2\,\mathrm{comp}(J[R])$. An edge that leaves R ends in U_{a-1}, in $U_{a'+1}$, or in one of the vertices $w_{a-1}, \dots, w_{a'+1}$, each of which has at most one further edge. So $|\delta R| < 6\Theta + \sigma + 3$, and $|R| < 10\Theta + \sigma + 1$. The cycle Z has a vertex in layer a and a vertex in layer a', so $|Z| \ge 2\sigma$. Also $|Z| \le |R|$. Hence $\sigma \le 10\Theta$ and $|R| \le 20\Theta$.

Step 3: the rank of J. Start with the interval of all layers. As long as J_I contains a cycle for some interval I of the present family, choose Z and R as in Step 2, and replace I by the at most two intervals that remain when $[a, a']$ is removed. Let $Z_1, \dots, Z_k$ be the chosen cycles and $R_1, \dots, R_k$ the corresponding sets. The sets R_i are pairwise disjoint, $|R_i| \le 20\Theta$, and $J - \bigcup_i R_i$ is a forest, because an edge of J joins equal or consecutive layers. So $\beta(J) \le 2\sum_i |R_i| \le 40\Theta k$.

The sets $V(Z_i)$ are pairwise disjoint cyclic regions of N that avoid P. Each has at most $|Z_i| \le 20\Theta$ edges that leave it. If $\xi \cap E(N - P)$ is a forest, then $\xi \cap E(N[V(Z_i)])$ is a forest for every i. So Lemma 2.5 and (5.2) give $k \le (20\Theta + 1)2^{20\Theta - 1}/\varepsilon$. Hence $\beta(J) \le C$ with

$$C = C_\varepsilon := \frac{40\Theta\,(20\Theta + 1)\,2^{20\Theta - 1}}{\varepsilon}.$$

Step 4: counting. The layers $L_0, \dots, L_\ell$ contain at least 2ℓ vertices, so $\ell \le b$. The vertices of A meet the ℓ edges of A and at most $\ell + 1$ other edges. If J is not empty, then $|E(J)| \ge (3b - 1) - (2\ell + 1)$ and $|V(J)| = 2b - \ell - 1$, so $\beta(J) \ge |E(J)| - |V(J)| + 1 \ge b - \ell$. If J is empty, then $2b = \ell + 1$. In both cases $\Delta := b - \ell \le C$. This is (i).

Compare N with a multigraph that has one vertex in L_0 and in L_ℓ, two in each layer between, and none beyond. Then N has exactly 2Δ vertices more. So at most 2Δ layers between L_0 and L_ℓ are not thin, and at most 2Δ layers lie beyond L_ℓ. This is (ii). A layer between L_0 and L_ℓ with $w \ge 3$ vertices has $w - 2 \ge w/3$ vertices too many. So at most $2 + 3 \cdot 2\Delta$ vertices lie outside the thin layers.

Finally let $y \in J$ be on a pair of parallel edges. The other end of the pair has two edges to y, so it is not on A, whose vertices have at most one further edge. So the pair is a cycle of J. Two such pairs have no common vertex, because N is subcubic and has more than two vertices. So there are at most $\beta(J) \le C$ pairs, and at most $2C$ thin layers are not ordinary. They contain at most $4C$ vertices. This gives (iii). ∎

Theorem 5.4 (Few cycles). For every $\varepsilon \in (0, \frac{1}{2}]$ there is a constant K_ε with the following property. Let N be a pole of rank b, and let s, t be two of its ports with $P_{st}(N) \geq (\frac{1}{2} + \varepsilon)2^b$. Then

$$c(N) \leq K_\varepsilon 2^{b/2}.$$

Proof. Let $C = C_\varepsilon$ be as in Proposition 5.3. We prove that a bridgeless 2-pole with these properties has at most $2^{31C+22} 2^{b/2}$ cycles. The theorem follows with $K_\varepsilon = 2^{31C+23}$. Indeed, in the notation of Lemma 5.2, each N_i has the required number of paths, and $\sum_i 2^{b_i/2} \leq 2^{b/2} + q - 1 \leq 2 \cdot 2^{b/2}$, because $x + y \leq xy + 1$ for $x, y \geq 1$.

So let N be a bridgeless 2-pole. If $b = 1$, then N has one cycle. Let $b \geq 2$, and use the notation above. Let Z be a cycle, and let L_i and L_j be the first and the last layer that it meets. Let M be the set of the vertices that lie in L_i, in L_j, or outside the ordinary layers. By Proposition 5.3(iii), $|M| \leq 10C + 6$.

Two strands. Let $L_l = \{w_l, y_l\}$ be an ordinary layer with $i < l < j$. The cycle Z has vertices before and after this layer. So it has at least two edges from L_{l-1} to L_l and at least two edges from L_l to L_{l+1}. The two vertices of L_l have degree at most two in Z. So Z has exactly two edges on each side of L_l and does not contain an edge $w_l y_l$. Suppose that both edges from L_{l-1} end in the same vertex of L_l. Then both edges to L_{l+1} start in the other vertex, and no edge of Z joins the part of Z before L_l to the part after it. This is impossible, because Z is connected. So each vertex of L_l has one edge of Z to L_{l-1} and one to L_{l+1}.

Matchings. Let L_l and L_{l+1} be ordinary layers with $i < l$ and $l + 1 < j$. By the last paragraph, the edges of Z between them form a perfect matching between $\{w_l, y_l\}$ and $\{w_{l+1}, y_{l+1}\}$. We count the perfect matchings. The edge $y_l y_{l+1}$ is not doubled, because the layers are ordinary. The edges $w_l y_{l+1}$ and $y_l w_{l+1}$ are not doubled, because a vertex of A has at most one further edge. If the edge $w_l w_{l+1}$ of A is doubled, then w_l and w_{l+1} have no edge to J. So there are at most two perfect matchings. If there are two, then w_{l+1} has two edge ends towards L_l: either both w_l and y_l are its neighbours, or $w_l w_{l+1}$ is doubled. In the same way w_{l+1} has two edge ends towards L_{l+2} if there are two perfect matchings between L_{l+1} and L_{l+2}. Both cannot happen, because w_{l+1} has degree three. So among consecutive pairs of ordinary layers, no two neighbouring pairs have two perfect matchings each.

The count. An edge of Z has an end in M, or it joins two consecutive ordinary layers strictly between L_i and L_j. So Z is determined by i and j, by its edges at M, and by its perfect matchings. There are at most $2^{3|M|}$ sets of edges at M. By the last paragraph there are at most $2^{(j-i)/2}$ choices for the matchings. All layers have an index of at most $n := b + C$, by Proposition 5.3(ii). Hence

$$c(N) \leq 2^{30C+18} \sum_{0 \leq i \leq j \leq n} 2^{(j-i)/2} \leq 2^{30C+18} \cdot 12 \cdot 2^{(b+C)/2} \leq 2^{31C+22} 2^{b/2}. \qquad \blacksquare$$

A pole with many paths also has a short cycle:

Lemma 5.5 (Four thin layers). Let N be a bridgeless 2-pole of rank at least two. If four consecutive layers are thin, then N has a cycle of length at most four all of whose vertices lie in these four layers.

Proof. The eight vertices of the four layers have degree three, and each has a neighbour in the layer before its own. The edges between two consecutive layers form a cut of N, so there are at least two of them. There are at most four: the two vertices of the earlier layer have six edge ends, and two of these point backwards. A pair of parallel edges among the eight vertices is a cycle of length two. Suppose there is none. If some cut between two of the four layers has four edges, it is a cycle of length four. So let the three cuts have $e_1, e_2, e_3 \in \{2, 3\}$ edges. Let r_2 and r_3 be the numbers of edges inside the second and the third layer. Counting edge ends in these layers gives $e_1 + e_2 + 2r_2 = 6$ and $e_2 + e_3 + 2r_3 = 6$. So $e_1 = e_2 = e_3$.

If all three are 2, then $r_2 = r_3 = 1$, and each vertex of the second and third layer has one edge to each side. The two edges between the second and the third layer and the edges inside these layers form a cycle of length four. If all three are 3, then $r_2 = 0$. In the second layer one vertex u has two neighbours in the first layer and the other vertex u' has one, say x; then x is also a neighbour of u. So u has one neighbour y in the third layer and u' has two, among them y. Then $xuyu'$ is a cycle of length four. ■

Corollary 5.6. For every $\varepsilon \in (0, \frac{1}{2}]$ and every integer $g \geq 0$ there is $b_1(\varepsilon, g)$ with the following property. Let G be a connected cubic multigraph without loops, of rank at least 9, that has no cyclic cut of size at most four. Let K be an induced subgraph of G that is a pole with at most $g + 2$ ports and rank $\beta(K) \geq b_1(\varepsilon, g)$. Then $P_{st}(K) < (\frac{1}{2} + \varepsilon)2^{\beta(K)}$ for all ports s, t of K.

Proof. Suppose that $P_{st}(K) \geq (\frac{1}{2} + \varepsilon)2^{\beta(K)}$, and let $N = K^\circ$ and $N_1, \ldots, N_q$ be as in Lemma 5.2. An edge of N is a path of K whose inner vertices are ports of K. Call the edge *plain* if this path has no inner vertex. At most g edges of N are not plain.

Let $1 < i < q$, and suppose that all edges of N_i are plain. Then the vertices of N_i have degree three in K. So exactly two edges of G leave $V(N_i)$, one for each bridge at N_i. The multigraph N_i contains a cycle, and so does N_1. This is a cyclic cut of size two, which G does not have. So $q \leq g + 2$, and some N_i has rank $b' \geq \beta(K)/(g + 2)$.

Let $C = C_\varepsilon$. By Proposition 5.3, applied to N_i, the distance between its ports is at least $b' - C$, and at most $2C$ of its inner layers are not thin. At most $2g$ layers contain an end of an edge that is not plain. Divide the inner layers into groups of four consecutive layers. If $b' \geq 9C + 8g + 9$, then in some group all layers are thin and all edges with both ends in the group are plain. Lemma 5.5 gives a cycle Z of length at most four in this group. It is a cycle of G. At most four edges of G leave $V(Z)$, and at most eight edges of G have an end in $V(Z)$. So $G - V(Z)$ has rank at least $\beta(G) - 8 > 0$, and $\delta V(Z)$ is a cyclic cut of size at most four. This is a contradiction. So the corollary holds with $b_1(\varepsilon, g) = (g + 2)(9C + 8g + 9)$. ■

Consequences of Theorem 5.4. From now on fix

$$\varepsilon_0 = \tfrac{1}{1000}, \qquad K_0 = K_{\varepsilon_0}, \qquad \kappa(h) = 2K_0\, 2^{-h/2}.$$

For four distinct ports a, b, c, d of a pole N let $f_{ab|cd}(N)$ be the number of pairs of vertex-disjoint paths of N, one from a to b and one from c to d.

Lemma 5.7. Let N be a pole of rank h. In each statement below, the letters denote distinct ports of N. Write P_{xy} for $P_{xy}(N)$ and f for $f(N)$.

(a) If $P_{ab} \geq 0.501 \cdot 2^h$, then $c(N) \leq K_0\, 2^{h/2}$.

(b) If $P_{ac} + P_{bc} \geq 1.002 \cdot 2^h$, then $P_{ab} \leq \kappa(h)\, 2^h$.

(c) If $P_{ac} + P_{ad} + P_{bc} + P_{bd} \geq 2.004 \cdot 2^h$, then $f_{ac|bd} + f_{ad|bc} \leq \kappa(h)\, 2^h$.

(d) If $P_{ab} + f_{ac|bd} + f_{ad|bc} \geq 1.002 \cdot 2^h$, then $P_{cd} \leq \kappa(h)\, 2^h$.

Proof. (a) is Theorem 5.4. In each of the other cases we change N into a pole N' and apply Theorem 5.4 to two ports of N'.

(b) Add a new vertex z and join it to a and b. Then N' has rank $h+1$, and $P_{zc}(N') = P_{ac} + P_{bc} \geq (\frac{1}{2} + \varepsilon_0)2^{h+1}$. The cycles of N' through z correspond to the paths of N from a to b. So $P_{ab} \leq c(N') \leq K_0\, 2^{(h+1)/2}$.

(c) Add a new vertex z joined to a and b, and a new vertex w joined to c and d. Then N' has rank $h+2$, and $P_{zw}(N')$ is the sum of the four path counts. The cycles of N' through z and w correspond to the pairs counted by $f_{ac|bd} + f_{ad|bc}$. So this sum is at most $K_0\, 2^{(h+2)/2}$.

(d) Add an edge between c and d. Then N' has rank $h+1$. A path from a to b in N' avoids the new edge, or it consists of the new edge and a pair counted by $f_{ac|bd} + f_{ad|bc}$. So $P_{ab}(N') = P_{ab} + f_{ac|bd} + f_{ad|bc}$. The cycles of N' through the new edge correspond to the paths of N from c to d. So $P_{cd} \leq K_0\, 2^{(h+1)/2}$. ∎

5.3 Full pairs

A *square* is a copy of the complete bipartite graph $K_{2,2}$; its two classes are its *left pair* and its *right pair*. A *single rung* is one edge; its two ends form both its left pair and its right pair. These are the two kinds of *tiles*. The *weight* of a square is 2, and the weight of a single rung is 1. A *chain* of tiles $\tau_1, \ldots, \tau_m$ consists of these tiles, pairwise disjoint, and of two disjoint edges from the right pair of τ_i to the left pair of τ_{i+1} for $1 \leq i < m$. These two edges are an *interface*. A *capped ladder* is a chain with two more vertices s and t, the *caps*: s is joined to the two vertices of the left pair of τ_1, and t to the two vertices of the right pair of τ_m (Figure 6). A capped ladder is a 2-pole. If its tiles have total weight W, it has $2W+2$ vertices and rank $W+1$.

Lemma 5.8 (Ladders). Let N be a bridgeless 2-pole whose pair of ports is full. Then N consists of two vertices and two parallel edges, or N is a capped ladder whose caps are the ports.

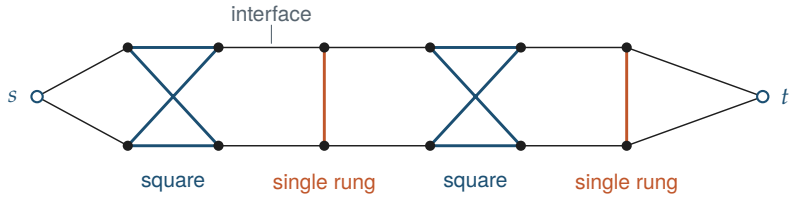


Figure 6. A capped ladder with four tiles, of weight 6 and rank 7. Without s and t it is a chain. If the right pair of the last tile is joined to the left pair of the first tile instead, the result is a ring (Section 6.1).

Proof. Let s, t be the ports. By Lemma 5.1, every cycle meets every path from s to t. If the rank is one, N is a cycle with two vertices. Otherwise N has no cut vertex, so there are two paths Q and Q' from s to t without a common inner vertex. Let F be the set of the edges outside $Q \cup Q'$. In F the inner vertices of Q and Q' have degree one, the other vertices except s, t have degree three, and s, t have degree zero.

A cycle in F would avoid Q. So F is a forest. Suppose that a component of F has two leaves on Q. The path of F between them and a part of Q form a cycle that avoids Q'. This is impossible, and the same holds for Q'. So every component of F has at most two leaves. A tree with degrees one and three has two more leaves than vertices of degree three. So every component of F with an edge is a single edge, from an inner vertex of Q to an inner vertex of Q'. Hence all vertices lie on $Q \cup Q'$, and F is a perfect matching between the inner vertices $x_1, \dots, x_k$ of Q and $y_1, \dots, y_k$ of Q', in the order from s to t. Write its edges as $x_i y_{\pi(i)}$.

Let $i < j < l$. If $\pi(l) < \min\{\pi(i), \pi(j)\}$, then the matching edges at x_i and x_j and the parts of Q and Q' between their ends form a cycle. The path that follows Q' from s to $y_{\pi(l)}$, crosses to x_l and follows Q to t avoids it. If $\pi(i) > \max\{\pi(j), \pi(l)\}$, the same holds for the cycle through the matching edges at x_j and x_l and the path that follows Q to x_i, crosses, and follows Q' to t. So neither case occurs. It follows that $\pi(1) \le 2$, and that $\pi(2) = 1$ if $\pi(1) = 2$. By induction, π is composed of fixed points and of exchanges of two consecutive numbers. A fixed point i is a single rung with the ends x_i, y_i. An exchange of i and $i+1$ is a square with the left pair x_i, y_i and the right pair x_{i+1}, y_{i+1}. The remaining edges of Q and Q' are the interfaces and the edges at the caps. ■

We need two counts in a capped ladder. Let $\tau_1, \dots, \tau_m$ be its tiles. First, let $1 \le i \le m$ and let u be a vertex of the left pair of τ_i. The number of paths from u to t inside $\tau_i \cup \dots \cup \tau_m \cup \{t\}$ is 2^q, where q is the weight of $\tau_i, \dots, \tau_m$. Indeed, such a path passes the tiles in order. It enters a tile at a vertex of the left pair and leaves it at a vertex of the right pair. In a square there are two paths to each vertex of the right pair. In a single rung the path stays or crosses. Second, let A_i be the number of paths between the two vertices of the right pair of τ_i inside $\{s\} \cup \tau_1 \cup \dots \cup \tau_i$, and $A_0 = 1$. This multigraph has rank equal to the weight p of $\tau_1, \dots, \tau_i$. So $A_i \le 2^p$ by (5.1).

Lemma 5.9 (A third port). Let T be a 3-pole of rank h with ports s, t, v. If the pair s, t is full, then $P_{vt}(T) = 2^h$ or $P_{vt}(T) \le \frac{7}{8} 2^h$.

Proof. By Lemma 5.2, T arises from a 2-pole N with ports s, t when an edge e is subdivided by v, and N consists of bridges and of bridgeless 2-poles $N_1, \dots, N_q$ with full pairs. If e is a bridge or an edge of some N_i with $i \geq 2$, then no path from v to t meets N_1. Such a path lies in a part of T of rank at most $h-1$, so $P_{vt}(T) \leq 2^{h-1}$ by (5.1). Let e be an edge of N_1, and let t_1 be the second port of N_1. A path from v to t is a path from v to t_1 followed by paths through $N_2, \dots, N_q$, and each of these 2-poles is full. So we may assume that $q = 1$.

By Lemma 5.8, N is a cycle with two vertices or a capped ladder. In the first case T is a triangle and $P_{vt} = 2 = 2^h$. In the second case let p be the weight of the tiles before e and q' the weight of the tiles after e; if e lies in a tile, this tile is not counted. A path from v to t never returns once it has entered the tiles after e, and from its point of entry it has $2^{q'}$ continuations. There are three cases.

The edge e is in an interface or at a cap. Then $h = p + q' + 1$. The path goes from v to the right, or it goes to the left, follows one of the A paths between the two ends of the interface on the left side, and uses the other edge of the interface. Here $A = A_i$ for the tile τ_i before e. So $P_{vt} = (1 + A)2^{q'}$. If $p = 0$, this is 2^h. If $p \geq 1$, then $A \leq 2^p$ and $P_{vt} \leq \frac{3}{4}\, 2^h$.

The edge e is a single rung. Then $h = p + q' + 2$. From each end of the rung the path continues to the right, or it makes the detour through the tiles before the rung to the other end of the rung and continues from there. So $P_{vt} = 2(1 + A)2^{q'}$, and the conclusion is the same.

The edge e is in a square. Then $h = p + q' + 3$. Let u, u' be the left pair and x, x' the right pair of the square, and let $e = ux$. A path from v leaves the square at x or x'. Without a detour through the tiles before the square, its part in the square is one of

$$vx, \qquad vux'u'x, \qquad vux', \qquad vxu'x'.$$

With a detour between u and u' it is one of $vu \cdots u'x$, $vu \cdots u'x'$ and $vxu' \cdots ux'$. There are A detours. So $P_{vt} = (4 + 3A)2^{q'}$. If $p = 0$, then $A = 1$ and $P_{vt} = \frac{7}{8}\, 2^h$. If $p \geq 1$, then $P_{vt} \leq (4 + 3 \cdot 2^p)2^{q'} \leq \frac{5}{8}\, 2^h$. ∎

Theorem 5.10 (Gap). Let N be a pole of rank b, and let s, t be two of its ports. Then

$$P_{st}(N) = 2^b \qquad \text{or} \qquad P_{st}(N) \leq \tfrac{15}{16}\, 2^b.$$

Proof. We use induction on b. Let $P_{st}(N) > 2^{b-1}$. By Lemma 5.2 we may assume that N is a 2-pole. If N has a bridge, the claim holds for the multigraphs N_i of Lemma 5.2, which have smaller rank, and the path counts multiply. So let N be bridgeless. If $b = 1$, the pair is full. Let $b \geq 2$. If s and t are adjacent, then $P_{st}(N) \leq 1 + 2^{b-1} \leq \frac{3}{4}\, 2^b$, by (5.1) for N without that edge. Otherwise let x and y be the neighbours of s. They are distinct, because N has no bridge. The multigraph $T = N - s$ is connected, because N has no cut vertex. It is a 3-pole of rank $b - 1$ with ports x, y, t, and

$$P_{st}(N) = P_{xt}(T) + P_{yt}(T).$$

By induction each term is 2^{b-1} or at most $\frac{15}{16}2^{b-1}$. If neither pair is full, we are done. If the pair x, t is full, then $P_{yt}(T)$ is 2^{b-1} or at most $\frac{7}{8}\,2^{b-1}$, by Lemma 5.9. So $P_{st}(N)$ is 2^b or at most $\frac{15}{16}2^b$. ■

Corollary 5.11 (Four ports). Let N be a 4-pole with ports a, b, c, d such that the four pairs ac, ad, bc, bd are full. Then one of the following holds.

(i) N is a chain of tiles; a, b is the left pair of its first tile and c, d is the right pair of its last tile.

(ii) N has a bridge e such that one component of $N - e$ contains a, b and a cycle, and the other component contains c, d and a cycle.

Proof. Add a vertex z joined to a, b and a vertex w joined to c, d. The result N' is a 2-pole of rank $\beta(N) + 2$ with ports z, w, and $P_{zw}(N') = 4 \cdot 2^{\beta(N)}$. So the pair z, w is full. Let $N_1, \dots, N_q$ be as in Lemma 5.2. Each N_i has a full pair of ports, so Lemma 5.8 applies to it. The neighbours a, b of z are distinct, so N_1 is a capped ladder with the cap z, and a, b is the left pair of its first tile. In the same way N_q is a capped ladder with the cap w. If $q = 1$, this is (i). If $q \geq 2$, let e be the bridge at N_1. The multigraphs $N_1 - z$ and $N_q - w$ are connected, each contains a cycle, and they lie in different components of $N - e$. This is (ii). ■

6 Rings and cyclic cuts of size at most four

In this section G is a connected cubic multigraph without loops, of rank r.

IDEA | the results of this section

We count the cycles of the rings. Then we show that a graph of large rank with a cyclic cut of size at most four is a ring, or it has at most $2^{r-1} + O(2^{r/2})$ cycles (Proposition 6.3 and Theorem 6.8). The proof uses the gap of Section 5: a side of the cut has all possible paths between two ends, or a fixed proportion of them is missing.

Let X be a set of $n \geq 1$ vertices of G with $|\delta X| = d$. Counting edge ends gives $\beta(G[X]) = (n - d)/2 + \mathrm{comp}(G[X])$. So $G[X]$ contains a cycle if $d \leq 2$, and also if $d = 3$ and $n \geq 2$. Hence every cut of size at most two is cyclic, and a cut of size three is cyclic unless one of its sides is a single vertex.

6.1 Rings

A *ring* consists of pairwise disjoint tiles $\tau_1, \dots, \tau_m$ and of two disjoint edges from the right pair of τ_i to the left pair of τ_{i+1} for every i, with indices modulo m. These pairs of edges are the *interfaces*. Let the ring have a squares and s single rungs, and let $W = 2a + s$ be its weight. We assume that $W \geq 7$. Then the ring is a cubic graph with $2W$ vertices and rank $W + 1$. Put

$$E(a, s) = 2^a\left(2a + 4s + \tbinom{s}{2} + \tfrac{1}{2}\right) - 3a - 4s + \tbinom{s}{2}.$$

Then $2^{2q} + E(q, 0)$ and $2^{2q+1} + E(q, 1)$ are the numbers $H(2q + 1)$ and $H(2q + 2)$ of (1.1).

Proposition 6.1 (Rings). A ring with $a \geq 1$ squares and s single rungs, of weight $W \geq 7$, has at most $2^W + E(a, s)$ cycles. Equality holds if $s \leq 1$. A ring of weight $W \geq 7$ without squares has at most $2^W + W^2$ cycles.

Proof. Two interfaces together form a cut. So a cycle uses an odd number of edges of every interface, or an even number of edges of every interface.

One edge of every interface. There are 2^m ways to choose the edges. The cycle then enters each tile at one vertex of its left pair and leaves it at one vertex of its right pair. In a square there are two such paths. In a single rung there is one: the cycle stays at the vertex or crosses the rung. Every choice gives a cycle. So there are $2^{m+a} = 2^W$ cycles of this kind.

No interface. These cycles are the a squares.

Both edges of every interface. In a square the cycle then uses two disjoint edges, and there are two ways to choose them. In a single rung it does not use the rung. Each of the 2^a choices is a cycle or a union of two cycles. Changing the choice in one square changes one into the other. So there are 2^{a-1} such cycles if $a \geq 1$, and at most one if $a = 0$.

Both edges of some interfaces, but not of all. Let the cycle use the interface after τ_i and not the interface before τ_i. Then it turns in τ_i: it joins the two vertices of the right pair inside the tile. It passes the next tiles on two disjoint paths, until it turns in a tile τ_j. This is already a closed curve, so the cycle uses exactly the interfaces from τ_i forward to τ_j. In each tile from τ_i to τ_j it has two possibilities if the tile is a square, and one if it is a single rung, in either role. Every choice gives a cycle. Let $I(i, j)$ be the set of the other tiles, those strictly after τ_j and strictly before τ_i. Then the number of these cycles is $2^a\, \omega(i, j)$, where $\omega(i, j) = 2^{-n}$ and n is the number of squares in $I(i, j)$. Every cycle of the present kind arises for exactly one ordered pair (i, j) of distinct tiles.

So a ring with $a \geq 1$ has $2^W + a + 2^{a-1} + 2^a\, \Omega$ cycles, where Ω is the sum of $\omega(i, j)$ over the ordered pairs of distinct tiles. We compute Ω. Let τ_j be a square. The squares $\tau_i \neq \tau_j$ give $\omega = 1, \frac{1}{2}, \ldots, 2^{2-a}$, in the order in which they follow τ_j. So the pairs of two squares contribute $a(2 - 2^{2-a})$. Let τ_i be a single rung. The squares τ_j give $\omega = 1, \frac{1}{2}, \ldots, 2^{1-a}$, in the order in which they precede τ_i. The same holds with the roles of i and j exchanged. So the pairs of a single rung and a square contribute $2s(2 - 2^{1-a})$. Finally let τ_i, τ_j be two single rungs, with n squares on one side between them and $a - n$ on the other. The two ordered pairs contribute $2^{-n} + 2^{-(a-n)} \leq 1 + 2^{-a}$, with equality if and only if $n = 0$ or $n = a$. Hence

$$\Omega \;\leq\; 2a + 4s + \tbinom{s}{2} - 2^{-a}\Big(4a + 4s - \tbinom{s}{2}\Big),$$

with equality if $s \leq 1$, because then there is no pair of two single rungs. This is the first statement.

If $a = 0$, every ordered pair of distinct tiles gives one cycle. There are $W(W - 1)$ pairs, and at most one further cycle uses both edges of all interfaces. ∎

Corollary 6.2. Let $r \geq 8$. Every ring of rank r has at most $H(r)$ cycles. A ring of rank r with $\lfloor (r-1)/2 \rfloor$ squares and at most one single rung has exactly $H(r)$ cycles.

Proof. Let $W = r - 1$. A ring of weight W with $a = \lfloor W/2 \rfloor$ squares has $s = W - 2a \leq 1$ single rungs, and it has $2^W + E(a, s) = H(r)$ cycles by Proposition 6.1. For $a \geq 2$ and $2a + s = W$, a direct computation gives

$$E(a-1, s+2) - E(a, s) = -2^{a-1}\Big(W + \frac{s^2+s}{2} - \frac{13}{2}\Big) + 2s - 4 \;<\; 0,$$

because $W \geq 7$. So $2^W + E(a, s) \leq H(r)$ for every ring of weight W with a square. The number $H(r) - 2^W$ is 71 for $W = 7$ and 124 for $W = 8$, and it more than doubles when W grows by two. So it exceeds W^2, and a ring without squares has fewer than $H(r)$ cycles. ■

We write R_r for a ring of rank $r \geq 8$ with $\lfloor (r-1)/2 \rfloor$ squares and at most one single rung. In the notation of Horak and Kim [5] it is the graph MC_{r-1}, and they computed $c(R_r) = H(r)$.

6.2 Cuts of size at most three

Proposition 6.3. There is a constant r_3 with the following property. Let G be a connected cubic multigraph without loops, of rank $r \geq r_3$, that has a cyclic cut of size at most three. Then

$$c(G) \;\leq\; 2^{r-1} + 5K_0^2\, 2^{r/2}.$$

Proof. Let $d \leq 3$ be the least size of a cyclic cut, and let δX be a cyclic cut of size d. By the remark at the start of this section, G has no cut of size less than d. So $G[X]$ and $G - X$ are connected, and no vertex is on two edges of δX if $d \geq 2$: otherwise we could move that vertex to the other side and get a smaller cut. Let T_1 and T_2 be the two sides, with ranks $b_1 \leq b_2$. Then $b_1 \geq 1$ and $r = b_1 + b_2 + d - 1$.

If $d = 1$, then $c(G) = c(T_1) + c(T_2) \leq (2^{b_1} - 1) + (2^{b_2} - 1) \leq 2^{r-1}$, because $b_1 + b_2 = r$.

Let $d \in \{2, 3\}$. Then T_1 and T_2 are d-poles, and the cut edges match their ports. A cycle that crosses the cut uses exactly two cut edges, and it consists of these and of one path in each side. For a pair j of cut edges let $x_j\, 2^{b_1}$ and $a_j\, 2^{b_2}$ be the numbers of paths between the corresponding ports of T_1 and of T_2. Then $0 \leq x_j, a_j \leq 1$ by (5.1), and

$$c(G) \;=\; c(T_1) + c(T_2) + 2^{b_1+b_2} \sum_j x_j a_j, \qquad 2^{r-1} = 2^{d-2}\, 2^{b_1+b_2}. \tag{6.1}$$

There is one pair for $d = 2$, and there are three pairs for $d = 3$. Let r be large. Then $b_2 \geq (r-2)/2$ is large, and $\kappa(b_2) \leq \frac{1}{1000}$. We have $c(T_1) < 2^{b_1}$.

Case A: $a_j \leq 0.501$ **for all** j. By Corollary 3.11, $c(T_2) \leq \frac{9}{16} 2^{b_2} \leq \frac{9}{32} 2^{b_1+b_2}$. So $c(G)\, 2^{-b_1-b_2}$ is less than $2^{-b_2} + \frac{9}{32} + 0.501$ for $d = 2$, and less than $2^{-b_2} + \frac{9}{32} + 1.503$ for $d = 3$. In both cases $c(G) < 2^{r-1}$.

Case B: $a_j > 0.501$ **for some** j. Then $c(T_2) \le K_0\, 2^{b_2/2}$ by Lemma 5.7(a). So $c(T_1) + c(T_2) < 2^{b_1} + K_0\, 2^{b_2/2} \le \frac{1}{1000} 2^{b_1+b_2}$.

Let $d = 2$. If $x_1 < 0.501$, then $c(G) < 0.502 \cdot 2^{b_1+b_2} < 2^{r-1}$ by (6.1). Otherwise $c(T_1) \le K_0\, 2^{b_1/2}$ by Lemma 5.7(a), and

$$c(G) \;\le\; 2^{r-1} + K_0\big(2^{b_1/2} + 2^{b_2/2}\big) \;\le\; 2^{r-1} + 2K_0\, 2^{r/2}.$$

Let $d = 3$, and number the pairs so that $a_1 \ge a_2 \ge a_3$. Any two pairs of ports of a 3-pole have one port in common. If $a_1 + a_2 < 1.002$, then $\sum_j x_j a_j < 1.503$ and $c(G) < 2^{r-1}$. Otherwise $a_3 \le \kappa(b_2)$ by Lemma 5.7(b). If $x_1 + x_2 < 1.002$, then $\sum_j x_j a_j < 1.003$ and $c(G) < 2^{r-1}$. Otherwise $x_3 \le \kappa(b_1)$ and $c(T_1) \le K_0\, 2^{b_1/2}$ by Lemma 5.7(b) and (a). Then (6.1) gives

$$c(G) \;\le\; 2K_0\, 2^{b_2/2} + 2^{b_1+b_2}\big(2+\kappa(b_1)\kappa(b_2)\big) \;=\; 2^{r-1} + 2K_0\, 2^{b_2/2} + 4K_0^2\, 2^{(b_1+b_2)/2} \;\le\; 2^{r-1} + 5K_0^2\, 2^{r/2}.$$

■

6.3 States of a pole with four ports

Let Y be a 4-pole of rank h with ports $1, 2, 3, 4$. A *pairing* is one of the three partitions $12|34$, $13|24$, $14|23$ of the ports into two pairs; the two pairs of a pairing are *opposite*. For a pair ij and a pairing $\alpha = ij|kl$ put

$$a_{ij} = 2^{-h} P_{ij}(Y), \qquad u_\alpha = 2^{-h} f_{ij|kl}(Y), \qquad S = \sum_{ij} a_{ij}, \qquad F = \sum_{\alpha} u_\alpha.$$

A pair of disjoint paths that joins the four ports is a join for the set of all four ports, and there are 2^h such joins. So $a_{ij} \le 1$ and $F \le 1$. We write $a_{ij}(Y)$, $S(Y)$ and so on when several poles are present. In this notation Lemma 5.7 reads as follows, for distinct ports i, j, k, l and with $\kappa = \kappa(h)$:

$$a_{ik} + a_{jk} \ge 1.002 \implies a_{ij} \le \kappa; \tag{6.2}$$

$$a_{ik} + a_{il} + a_{jk} + a_{jl} \ge 2.004 \implies u_{ik|jl} + u_{il|jk} \le \kappa; \tag{6.3}$$

$$a_{kl} > \kappa \implies a_{ij} + F - u_{ij|kl} < 1.002. \tag{6.4}$$

Fix h_0 such that $\kappa(h) \le \frac{1}{1000}$ for $h \ge h_0$.

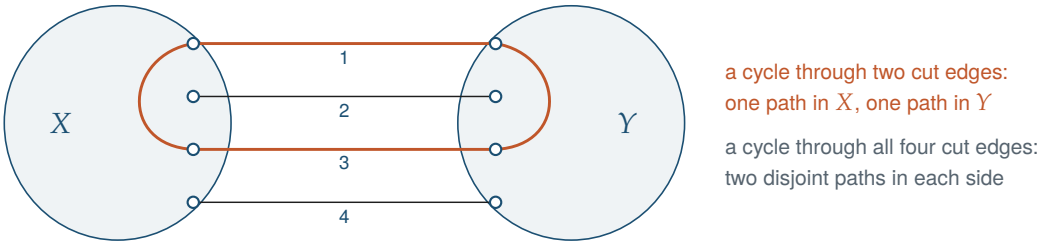


Figure 7. Two 4-poles X and Y, joined at their ports. The number a_{ij} counts the paths between the ports i and j of one side, and u_α counts the pairs of disjoint paths for a pairing α of the ports, both as a proportion of the largest possible number.

Lemma 6.4 (Gluing). Let X and Y be 4-poles of ranks h and k, and let G arise from them when we join each port of X to the port of Y with the same number. Then $\beta(G) = h+k+3$ and

$$c(G) = c(X) + c(Y) + 2^{h+k}\,\mathcal{B}, \qquad \mathcal{B} = \sum_{ij} a_{ij}(X)\,a_{ij}(Y) + \sum_{\alpha\neq\beta} u_\alpha(X)\,u_\beta(Y) \;\le\; S(Y) + F(Y).$$

Proof. A cycle that crosses the cut uses two or four of the new edges. With two, at the ports i and j, it consists of a path between these ports in each pole. With four, it consists of a pair of disjoint paths in each pole. If the pairings in X and Y are equal, the union is two cycles. If they differ, it is one cycle. The inequality holds because $a_{ij}(X) \le 1$ and $F(X) \le 1$. ∎

Lemma 6.5 (Low states). Let Y be a 4-pole of rank $h \ge h_0$ with $a_{ij} \le 0.51$ for all six pairs. Then $S + F \le 3.82$, and $S + \frac{1}{2}F + \frac{1}{2}u_\gamma \le 3.7$ for every pairing γ.

Proof. For a pairing $\alpha = ij|kl$ let $M_\alpha = \max\{a_{ij}, a_{kl}\}$. If both numbers are larger than κ, then $M_\alpha + F - u_\alpha < 1.002$ by (6.4).

If $S \le 2.75$, the first claim holds because $F \le 1$. Otherwise every a_{ij} is at least $S - 5\cdot 0.51 > 0.2$. Adding the three inequalities gives $\sum_\alpha M_\alpha + 2F < 3.006$. As $S \le 2\sum_\alpha M_\alpha$ and $S \le 3.06$, we get $S + F < 1.503 + \frac{3}{4}S \le 3.8$.

If $S \le 2.7$, the second claim holds because $u_\gamma \le F \le 1$. Otherwise every a_{ij} is at least 0.15. Adding the inequalities for the two pairings α, β other than γ gives $M_\alpha + M_\beta + F + u_\gamma < 2.004$. Hence $S + \frac{1}{2}F + \frac{1}{2}u_\gamma < S - \frac{1}{2}M_\alpha - \frac{1}{2}M_\beta + 1.002$. In this bound the two pairs of α contribute at most $\frac{3}{2}\cdot 0.51$, and so do the two pairs of β. The two pairs of γ contribute at most $2\cdot 0.51$. The total is at most $2.55 + 1.002$. ∎

Lemma 6.6 (Rigidity). Let Y be a 4-pole of rank $h \ge h_0$ with $S + F \ge 3.99$. Then three pairs with a common port are full, or the four pairs between two disjoint pairs of ports are full.

Proof. For a partition of the ports into two pairs, the *cross sum* is the sum of a_{ij} over the four pairs ij that are not pairs of the partition. Each pair ij occurs in two of the three cross sums. We use Theorem 5.10 in this form: if $a_{ij} > \frac{15}{16}$, then ij is full.

Case 1: some cross sum is at least 2.004. Let it be the cross sum X of $12|34$. Then $a_{13}+a_{23} \ge 1.002$ or $a_{14}+a_{24} \ge 1.002$, so $a_{12} \le \kappa$ by (6.2). In the same way $a_{34} \le \kappa$. By (6.3), $u_{13|24} + u_{14|23} \le \kappa$. Put $u = u_{12|34}$. Then

$$S + F \;\le\; X + u + 3\kappa.$$

At most one of the four numbers in X is at most κ, because $X \ge 2.004$. Let ij be one of the four pairs, and kl the opposite pair. If $a_{kl} > \kappa$, then (6.4) gives $u < 1.002 - a_{ij}$.

Suppose that all four numbers are larger than κ. Then $u < 1.002 - m$, where m is the largest of them. So $S+F$ is less than the sum of the three smallest of them plus 1.005. As $S+F \geq 3.99$, each of the three is larger than 0.985. So all four pairs are full.

Suppose that one of the four numbers is at most κ, say a_{24}. Then $u < 1.002 - \max\{a_{14}, a_{23}\}$, and $S + F < a_{13} + \min\{a_{14}, a_{23}\} + 1.002 + 4\kappa \leq 3.006$. This contradicts $S + F \geq 3.99$.

Case 2: all cross sums are less than 2.004. Their sum is $2S$, so $S < 3.006$. Hence $F > 0.984$ and $S \geq 2.99$, and the sum of two opposite numbers is more than $2.99 - 2.004 = 0.986$.

Suppose that all six numbers are larger than κ. For a pairing $\alpha = ij|kl$, (6.4) gives $\max\{a_{ij}, a_{kl}\} + F - u_\alpha < 1.002$, and the maximum is at least 0.493. So $F - u_\alpha < 0.509$. Adding this for the three pairings gives $2F < 1.527$, a contradiction.

So one number is at most κ, say a_{34}. Then $a_{12} > 0.985$, so 12 is full. By (6.2), $a_{13} < 0.002$ or $a_{23} \leq \kappa$; also $a_{14} < 0.002$ or $a_{24} \leq \kappa$. If a number is less than 0.002, its opposite is more than 0.984, hence full. If $a_{13} < 0.002$, then 24 is full, so $a_{14} < 0.002$, and 23 is full. Then the pairs $12, 23, 24$ are full. If $a_{23} < 0.002$, then 14 is full, so $a_{24} < 0.002$, and 13 is full. Then the pairs $12, 13, 14$ are full. ∎

Lemma 6.7 (Full star). Let Y be a 4-pole in which the pairs 12, 13 and 14 are full. Then $u_\alpha(Y) \leq \frac{1}{2}$ for every pairing α.

Proof. Let $\alpha = 1j|kl$, and let h be the rank of Y. Consider a path Q from 1 to j and a disjoint path Q' from k to l. The pair $1j$ is full, so every cycle of Y meets Q (Lemma 5.1). So $Y - V(Q)$ is a forest, and Q determines Q'. Hence $f_{1j|kl}(Y)$ is at most the number of paths from 1 to j in $Y - k$.

The port k lies on a cycle of Y. Otherwise both edges at k are bridges, and one of them has a side that contains neither k nor the port 1. This side contains a cycle, by the count of edge ends in the proof of Lemma 5.2, and no path from 1 to k meets it. This contradicts Lemma 5.1 for the full pair $1k$. So $Y - k$ is connected of rank $h - 1$, and it has at most 2^{h-1} paths from 1 to j by (5.1). ∎

6.4 Cuts of size four

IDEA | outline of the proof of Theorem 6.8

Let Y be the larger side of a cut of size four. In units of 2^{h+k}, the number of cycles across the cut is at most $S(Y) + F(Y)$, and $c(G) \geq 2^{r-1}$ requires the value 4. If no pair of ports of Y is close to full, then $S + F \leq 3.82$ by Lemma 6.5. In the other case Y has few cycles, and by Lemma 6.6 there are two possible shapes. If three pairs at one port are full, then $c(G) < 2^{r-1}$ by Lemma 6.7. If the four pairs between two pairs of ports are full in both sides, then both sides are chains of tiles, and G is a ring.

Theorem 6.8. There is a constant r_4 with the following property. Let G be a connected cubic multigraph without loops, of rank $r \geq r_4$, that has a cyclic cut of size four and no cyclic cut of smaller size. Then G is a ring, or $c(G) < 2^{r-1}$.

Proof. Let δX be a cyclic cut of size four. By the remark at the start of the section, G has no cut of size at most two, and every cut of size three has a single vertex on one side. So $G[X]$ and $G - X$ are connected. No vertex v is on two edges of δX: otherwise the cut around $X \setminus \{v\}$, or around $X \cup \{v\}$, would be a cyclic cut of size three. So the two sides are 4-poles. Call them X and Y, with ranks $h \leq k$, and number the ports so that Lemma 6.4 applies. Then $r = h + k + 3$, and k is large when r is large. We write κ for $\kappa(k)$ and assume that $k \geq h_0$. By Lemma 6.4,

$$\frac{c(G)}{2^{h+k}} = \frac{c(X)}{2^{h+k}} + \frac{c(Y)}{2^{h+k}} + \mathcal{B}, \qquad 2^{r-1} = 4 \cdot 2^{h+k}, \qquad c(X) < 2^h. \tag{6.5}$$

Case 1: $a_{ij}(Y) \leq 0.51$ for all pairs. By Corollary 3.11, $c(Y) \leq \frac{9}{16} 2^k$. If $h \geq 2$, then $\mathcal{B} \leq S(Y) + F(Y) \leq 3.82$ by Lemma 6.5, and the right side of (6.5) is less than $2^{-k} + \frac{9}{64} + 3.82 < 4$. If $h = 1$, then X has four vertices and is a cycle. Number its vertices $1, 2, 3, 4$ along the cycle. Then all $a_{ij}(X)$ are 1, and $u_{12|34}(X) = u_{14|23}(X) = \frac{1}{2}$ and $u_{13|24}(X) = 0$. So $\mathcal{B} = S(Y) + \frac{1}{2}F(Y) + \frac{1}{2}u_{13|24}(Y) \leq 3.7$ by Lemma 6.5, and the right side of (6.5) is less than $2^{-k} + \frac{9}{32} + 3.7 < 4$. In both cases $c(G) < 2^{r-1}$.

In the other cases $a_{ij}(Y) > 0.51$ for some pair, so $c(Y) \leq K_0\, 2^{k/2}$ by Lemma 5.7(a). The first two terms on the right of (6.5) are then less than $2^{-k} + K_0\, 2^{-k/2}$.

Case 2: $S(Y) + F(Y) < 3.99$. Then $\mathcal{B} < 3.99$ and $c(G) < 2^{r-1}$.

Case 3: $S(Y) + F(Y) \geq 3.99$, and the pairs $12, 13, 14$ are full in Y. By (6.2), the other three numbers $a_{ij}(Y)$ are at most κ. If one of the pairs $12, 13, 14$ is not full in X, then $a_{ij}(X) \leq \frac{15}{16}$ for this pair by Theorem 5.10. So $\mathcal{B} \leq 3 - \frac{1}{16} + 3\kappa + F(X)F(Y) < 4 - \frac{1}{32}$, and $c(G) < 2^{r-1}$.

Let the three pairs be full in X as well. Then the other three numbers $a_{ij}(X)$ are at most $\kappa(h)$, and $c(X) \leq K_0\, 2^{h/2}$, by Lemma 5.7. By Lemma 6.7, all numbers $u_\alpha(X)$ and $u_\alpha(Y)$ are at most $\frac{1}{2}$. Let v be the largest $u_\alpha(Y)$. The numbers $u_\alpha(X)$ are at most $\frac{1}{2}$ and add up to at most 1, so

$$\sum_{\alpha \neq \beta} u_\alpha(X)\, u_\beta(Y) = \sum_{\alpha} u_\alpha(X)\big(F(Y) - u_\alpha(Y)\big) \leq \frac{F(Y) + v}{2} \leq \frac{3}{4}.$$

Hence $\mathcal{B} \leq 3 + 3\kappa(h)\kappa(k) + \frac{3}{4}$, and the right side of (6.5) is less than 4 when k is large. So $c(G) < 2^{r-1}$.

Case 4: $S(Y) + F(Y) \geq 3.99$, and the pairs $13, 14, 23, 24$ are full in Y. By (6.2), (6.3) and (6.4), $a_{12}(Y)$ and $a_{34}(Y)$ are at most κ, and $F(Y) \leq 0.002 + \kappa$. If one of the four pairs is not full in X, then $\mathcal{B} \leq 4 - \frac{1}{16} + 2\kappa + F(Y) < 4 - \frac{1}{32}$, and $c(G) < 2^{r-1}$.

Otherwise the four pairs are full in X and in Y, and Corollary 5.11 applies to both. In its case (ii) the pole has a bridge e, and the component of the pole without e that contains the

ports $1, 2$ contains a cycle. Its vertex set has three edges of G that leave it: e and the two edges at the ports. The other side contains a cycle as well. This is a cyclic cut of size three, which G does not have. So X and Y are chains of tiles, from the pair $1, 2$ to the pair $3, 4$. The edges of the cut join the first tile of X to the first tile of Y and the last tile of X to the last tile of Y. So G is a ring.

By Lemma 6.6, these are all cases, up to the numbering of the ports. ■

7 Bounded girth

Theorem 7.1 (Bounded girth). For every integer $g_0 \geq 2$ there is $R(g_0)$ with the following property. Let G be a connected cubic multigraph without loops, of rank $r \geq R(g_0)$ and girth at most g_0, that has no cyclic cut of size at most four. Then $c(G) < 2^{r-1}$.

IDEA | outline of the proof

Sort the cycles through an edge e of a shortest cycle C by the vertex where they leave C. The cycles that leave after p steps are paths in a pole of rank at most $r - 1 - p$, and such a pole has about half of the possible paths at most. The sum over p is a geometric series with fewer terms than the length g of C. So it is smaller than the full series by a factor $1 - 2^{1-g}$, and this difference gives the bound.

Proof. Let $r \geq 9$, and let g be the girth of G. A cycle Z of length at most four would give the cyclic cut $\delta V(Z)$ of size at most four, as in the proof of Corollary 5.6. So $5 \leq g \leq g_0$. Let $C = v_0 v_1 \cdots v_{g-1}$ be a shortest cycle. It has no chord. A vertex outside C has at most one neighbour on C, because two neighbours would give a cycle of length at most $\lfloor g/2 \rfloor + 2 < g$. So each v_i has exactly one neighbour w_i outside C, the vertices w_i are distinct, and every vertex of $G - V(C)$ has degree at least two in $G - V(C)$. Hence every component of $G - V(C)$ contains a cycle.

Let e be the edge $v_0 v_{g-1}$, and put $N = G - e$ and $b = r - 1$. Then N is connected of rank b, and $c(G) = c(N) + P$, where P is the number of paths from v_0 to v_{g-1} in N. One of these paths follows C. Every other path leaves C for the first time at a vertex v_{p-1} with $1 \leq p \leq g - 1$, through the edge $v_{p-1} w_{p-1}$. Its remaining part is a path from w_{p-1} to v_{g-1} in $N - \{v_0, \ldots, v_{p-1}\}$. Let K_p be the component of v_{g-1} in this multigraph, and let P_p be the number of paths from w_{p-1} to v_{g-1} in K_p; put $P_p = 0$ if $w_{p-1} \notin K_p$. Then

$$P = 1 + \sum_{p=1}^{g-1} P_p.$$

The rank of K_p. We get $N - \{v_0, \ldots, v_{p-1}\}$ from N when we delete p vertices and $2p$ edges. So the ranks of its components add up to $b - p$ plus the number of components other than K_p. Each of these is a component of $G - V(C)$ and contains a cycle. Hence $\beta(K_p) \leq b - p$.

The shape of K_p. Let $p \le g-2$. Then K_p is an induced subgraph of G. Its vertices have degree three in K_p, except v_p, v_{g-1} and the vertices w_i with $i < p$, which have degree two. So K_p is a pole with at most g ports, and w_{p-1} and v_{g-1} are two of its ports if $P_p > 0$. For $p = g-1$, the vertex v_{g-1} has degree one in K_p. Delete it. The result is an induced subgraph of G of the same rank. It is a pole with ports among the w_i, and P_p is the number of its paths from w_{g-2} to w_{g-1}.

The count. Put $\varepsilon = 2^{-g-2}$, and let $b_1 = b_1(\varepsilon, g-2)$ be as in Corollary 5.6. If $\beta(K_p) \ge b_1$, the corollary gives $P_p < (\frac{1}{2}+\varepsilon)2^{\beta(K_p)} \le (\frac{1}{2}+\varepsilon)2^{b-p}$. If $\beta(K_p) < b_1$, then $P_p \le 2^{b_1}$ by (5.1). Hence

$$P \;\le\; 1 + g\,2^{b_1} + \Big(\frac{1}{2}+\varepsilon\Big)\sum_{p=1}^{g-1} 2^{b-p} \;\le\; 2^{b-1} - \frac{3}{4}\,2^{b-g} + g\,2^{b_1} + 1.$$

By Corollary 3.11, $c(N) \le \psi(r-1) \le 2^{r-2} + o(2^r)$. So

$$c(G) \;\le\; 2^{r-1} - \frac{3}{4}\,2^{r-1-g} + o(2^r) + g\,2^{b_1} + 1.$$

The numbers g and b_1 are bounded in terms of g_0. So $c(G) < 2^{r-1}$ when r is large. ■

8 Large girth outside a small set

Recall that $q(G,P)$ is the probability that $\xi \cap E(G-P)$ is a forest.

Theorem 8.1. There are integers $D_* \ge 7$ and r_5 with the following property. Let G be a connected cubic multigraph of rank $r \ge r_5$. Let P be a nonempty set of at most $3\log r$ vertices such that $G[P]$ is connected. If $G-P$ has girth at least D_*, then $q(G,P) < \frac{2}{5}$.

IDEA | outline of the proof

Suppose that $q(G,P) \ge \frac{2}{5}$. We find two disjoint parts of $G-P$ that are large, have small boundaries and have large girth. Each of them has a forest proportion of at most $\frac{3}{5}$, for every choice of the edges outside it. This is a contradiction, because $(\frac{3}{5})^2 < \frac{2}{5}$. To find the parts we use a packing bound for short cycles (Section 8.2), a second moment and a chain of nested separators (Section 8.3).

8.1 Pieces with a small boundary

Let N be a pole, and let T be a set of ports of even size. A *T-join* of N is a set of edges in which the vertices of T have odd degree and all other vertices have even degree. There are $2^{\beta(N)}$ of them. The *forest proportion* of T is the proportion of the T-joins that are forests. For $T = \emptyset$ the T-joins are the even subgraphs, only the empty one is a forest, and the forest proportion is $2^{-\beta(N)}$. Put

$$R_* = R_{1/20}, \qquad B = 2R_* + 2^{100} + 1,$$

with $R_{1/20}$ as in Proposition 3.9. Then $\Theta_{1/20}(k) + k \le B(k+1)^2$ for all $k \ge 1$.

Lemma 8.2 (A marker in a pole). Let N be a pole with $k \ge 2$ ports, and let o be a port. Call the set of the vertices at distance j from o the *layer* j. Suppose that some set T of ports has forest proportion at least $\frac{3}{5}$. Then each layer has at most $B(k+1)^2$ vertices.

Proof. Take a breadth-first tree of N from o, and let P be the union of its paths from o to the other ports. Then $N[P]$ is connected, P contains all vertices of degree two, and P has at most k vertices in each layer, because a path of the tree from o has one vertex in each layer that it meets. The tree paths inside P contain a T-join F_0. The map $J \mapsto J + F_0$ is a bijection from the T-joins of N to the even subgraphs of N, and $J + F_0$ has the same edges in $N - P$ as J. So if J is a forest, then $J + F_0$ is a forest in $N - P$. Hence $q(N, P) \ge \frac{3}{5} > \frac{1}{2} + \frac{1}{20}$. The distance from o is a level function. By Lemma 4.1 with $\delta = \frac{1}{20}$, each layer has fewer than $\Theta_{1/20}(k)$ vertices outside P, and it has at most k vertices in P. ∎

Lemma 8.3 (Girth and rank of a pole). Let N be a pole of rank $h \ge 2$ with k ports and girth at least D. Put $t = \lceil \log(8h) \rceil$. Then

$$D \;\le\; 4t\Big(1 + \frac{k}{3(h-1)}\Big).$$

Proof. Replace every vertex of degree two and its two edges by one edge, and give each edge of the result the length of the path of N that it replaces. The result N_0 is a connected cubic multigraph with $n_0 = 2h - 2$ vertices and $m_0 = 3h - 3$ edges, and the lengths of its edges add up to $m_0 + k$. A cycle of N_0 comes from a cycle of N of the same length.

There are $2m_0\, 2^{t-1}$ non-backtracking walks with t edges in N_0. Choose one of them uniformly. Then each of its steps is uniform over the $2m_0$ directed edges. So the expected length of the walk is $t(1 + k/m_0)$. At least half of the walks have length at most twice this value. Their number is at least

$$m_0\, 2^{t-1} \;\ge\; 12h(h-1) \;>\; n_0^2.$$

If one of these short walks visits a vertex twice, it contains a cycle that is not longer than the walk. Otherwise all of them are paths, and two of them have the same first vertex and the same last vertex. Their union contains a cycle of length at most the sum of their lengths. In both cases N has a cycle of length at most $4t(1 + k/m_0)$. ∎

Lemma 8.4 (Forest proportions at large girth). There are absolute constants $D_* \geq 7$ and C_0 with the following property. Let N be a pole of rank h with k ports and girth at least D. Suppose that

(a) $h \geq 256$, $k \leq 4D$ and $D \geq D_*$, or

(b) $h \geq \Lambda \geq 4$, $k \leq 2\Lambda$ and $D \geq C_0 \log(\Lambda + 2)$ for some number Λ.

Then every set of ports of even size has forest proportion at most $\frac{3}{5}$.

Proof. If $k \leq 1$, the only set is the empty one, with forest proportion 2^{-h}. Let $k \geq 2$, and put $t = \lceil \log(8h) \rceil \leq \log h + 4$.

The rank is exponential in the girth. In case (a), $h - 1 > \frac{32}{3}t$: this holds at $h = 256$, where $t = 11$, and the ratio $(h-1)/(\log h + 4)$ increases with h. So Lemma 8.3 gives $D \leq 4t + 16tD/(3(h-1)) < 4t + D/2$. In case (b), $k/(3(h-1)) \leq 2h/(3(h-1)) < 1$, and Lemma 8.3 gives $D < 8t$. In both cases $D < 8t \leq 8 \log h + 32$, so

$$h \geq 2^{D/8-4}, \qquad \text{and } N \text{ has } n \geq h \text{ vertices.} \tag{8.1}$$

A ball without ports. Put $s = \lfloor D/16 \rfloor$. A ball of radius s has at most $3 \cdot 2^s$ vertices. So at most $3k\,2^s$ vertices are within distance s of a port. In case (a) this is at most $12D\,2^{D/16}$, and in case (b) at most $6\Lambda\,2^{D/16}$. Choose D_* and C_0 so large that this number is less than $2^{D/8-4} \leq n$ in each case. Then some vertex w has distance more than s from every port. All vertices within distance s of w have degree three, and $2s < D$. So the ball of radius s around w is a tree in which every vertex at distance less than s from w has degree three, and it has at least $2^s \geq 2^{D/16-1}$ vertices. Fix a port o. The distances from o to two vertices of this ball differ by at most $2s$. So the ball meets at most $2s + 1 \leq D$ layers with respect to o, and some layer has at least $2^{D/16-1}/D$ vertices.

Conclusion. Suppose that some set of ports has forest proportion more than $\frac{3}{5}$. By Lemma 8.2 every layer has at most $B(k+1)^2$ vertices. This is at most $B(4D+1)^2$ in case (a) and at most $B(2\Lambda+1)^2$ in case (b). In case (a) choose D_* so large that $2^{D/16-1}/D > B(4D+1)^2$ for $D \geq D_*$. In case (b) the function $2^{D/16}/D$ increases for $D \geq 32$, and at $D = C_0 \log(\Lambda+2)$ its ratio to $(\Lambda+2)^2$ is $(\Lambda+2)^{C_0/16-2}/(C_0 \log(\Lambda+2))$. This tends to infinity with C_0, uniformly in $\Lambda \geq 4$. So a large C_0 gives $2^{D/16-1}/D > B(2\Lambda+1)^2$. In both cases we have a contradiction. ∎

Products of pieces.

Lemma 8.5 (Product). Let G be a cubic multigraph, let P be a set of vertices, and let $U_1, \ldots, U_m \subseteq V(G) \setminus P$ be pairwise disjoint sets such that each $G[U_i]$ is a pole. Suppose every even set of ports in each pole has forest proportion at most $3/5$. Then $q(G, P) \leq (3/5)^m$.

Proof. Let E' consist of the edges that lie in none of the induced poles. Fix a restriction $x \subseteq E'$ that extends to an even subgraph of G. For each i, let T_i be the vertices of U_i incident with an odd number of edges of x. The parity equations say that $\xi \cap E(G[U_i])$ is a T_i-join. These equations involve disjoint internal edge sets, so the joins are independent and uniform, conditional on $\xi \cap E' = x$. A vertex of degree three in a pole has no incident edge in E'; thus each T_i is an even set of ports. If $\xi \cap E(G-P)$ is a forest, its restriction to every pole is a forest. The conditional probability is therefore at most $(3/5)^m$. Average over x. Edges between the poles belong to E', so adjacent pieces cause no exception. ∎

Lemma 8.6 (The core of a piece)**.** Let G be a cubic multigraph, and let X be a set of vertices such that $G[X]$ is connected, has no loop, and has rank $h \geq 2$. Then there is a set $U \subseteq X$ such that $G[U]$ is a pole of rank h with at most $|\delta X|$ ports, and the girth of $G[U]$ is at least the girth of $G[X]$.

Proof. Delete vertices of degree one from $G[X]$ as long as there are any. The result $G[U]$ is connected, has the same cycles as $G[X]$ and all its degrees are two or three. When a vertex of degree one is deleted, two edges that left the set no longer do, and one edge inside the set now leaves it. So $|\delta U| \leq |\delta X|$. A vertex of degree two in $G[U]$ has exactly one edge end outside $G[U]$, and a vertex of degree three has none. So $G[U]$ has $|\delta U|$ ports. ∎

Proposition 8.7 (Pieces with small boundaries)**.** Let G be a connected cubic multigraph and let P be a set of vertices. Let $X_1, \ldots, X_m \subseteq V(G) \setminus P$ be pairwise disjoint, nonempty and proper sets of vertices. Suppose that one of the following holds for every i.

(a) $G[X_i]$ is connected of rank at least 256 and girth at least $D \geq D_*$, and $|\delta X_i| \leq 4D$.

(b) $|X_i| \geq 8\Lambda^2$, $|\delta X_i| \leq 2\Lambda$, and $G[X_i]$ has girth at least D, where $\Lambda \geq 4$ and $D \geq C_0 \log(\Lambda + 2)$.

Then $q(G, P) \leq (3/5)^m$.

Proof. It suffices to find sets $U_i \subseteq X_i$ as in Lemma 8.5. In case (a), Lemma 8.6 gives a pole $G[U_i]$ of rank at least 256 with at most $4D$ ports and girth at least D, and Lemma 8.4(a) applies.

In case (b) write $n = |X_i|$, $d = |\delta X_i|$ and $c = \mathrm{comp}(G[X_i])$. Every component of $G[X_i]$ has an edge that leaves X_i, so $1 \leq c \leq d \leq 2\Lambda$. Count the edge ends: $3n = 2|E(G[X_i])| + d$. So $\beta(G[X_i]) = (n-d)/2 + c$, and some component Y of $G[X_i]$ has rank at least

$$\frac{n-d}{2c} + 1 \;\geq\; \frac{8\Lambda^2 - 2\Lambda}{4\Lambda} + 1 \;>\; \Lambda.$$

Its boundary is a part of δX_i. Lemma 8.6 for Y gives a pole $G[U_i]$ of rank at least Λ with at most 2Λ ports and girth at least D, and Lemma 8.4(b) applies. ∎

8.2 Packing of short cycles

IDEA | the purpose of the packing bound

The second-moment argument requires a graph of girth about $\log r$. So we delete a maximal family of disjoint short cycles, and we need a bound for the size of this family. Lemma 2.5 gives such a bound only when the proportion of forests exceeds one half by a fixed margin. Here we prove the bound with Proposition 8.7. Many disjoint cyclic sets with small boundaries have independent local events, or many of them lie in one string (Lemma 8.8), and a long string contains two pieces to which Proposition 8.7 applies.

Throughout, $h_* = 256$.

Vertices that share a parity. Let Q be a connected multigraph without loops. For an edge e, the map $\xi \mapsto [e \in \xi]$ is a linear functional on $\mathcal{Z}(Q)$. For a vertex v let L_v be the space of functionals that is spanned by the edges at v. For a set A of vertices put $U_A = \sum_{a\in A} L_a$. We use one standard fact: if a set F of edges meets every even subgraph of Q in an even number of edges, then $F = \delta Y$ for some set Y of vertices. (Fix a spanning tree and a root. Let Y be the set of the vertices whose tree path from the root has an odd number of edges in F. Then F and δY contain the same tree edges, and the fundamental cycles show that they contain the same edges outside the tree.)

Lemma 8.8 (Strings)**.** Let Q be a connected multigraph without loops, and let A be a nonempty proper set of its vertices. Put $b = |\delta A|$. Call a vertex $v \notin A$ *relevant* if $L_v \cap U_A \neq 0$. There are a set of at most $2b$ relevant vertices, the *exceptions*, and at most $2b$ sequences of relevant vertices, the *strings*, with these properties. Every relevant vertex that is not an exception is in a string. If $v_1, \dots, v_m$ is a string and $1 \le i < j \le m$ with $j \ge i + 2$, there is a set $M(i, j)$ of vertices such that

(i) $M(i, j)$ is disjoint from $A \cup \{v_i, v_j\}$, the graph $Q[M(i, j)]$ is connected, and $v_l \in M(i, j)$ for $i < l < j$;

(ii) every edge that leaves $M(i, j)$ ends in v_i or in v_j;

(iii) $M(i, j) \cap M(i', j') = \emptyset$ if $j \le i'$.

Proof. Call the edges of δA *terminals*. The end of a terminal outside A is its *foot*. Every component of $Q - A$ contains a foot, because Q is connected.

Step 1. Let $v \notin A$ be relevant and not a foot, and let K be the component of $Q - A$ that contains v. We show that at least two components of $K - v$ contain feet. Let $0 \neq a \in L_v \cap U_A$. Then a is the sum of the functionals of a set F of edges at v, and also of a set F' of edges with an end in A. No edge is in both sets, because v is not a foot. The set $F \cup F'$ meets every even subgraph in an even number of edges, so $F \cup F' = \delta Y$ for some Y. We may assume that $v \notin Y$. No edge of $K - v$ is in $F \cup F'$. So each component of $K - v$ lies inside Y or outside Y, and F is the set of all edges from v to the components inside Y. For a component J of $K - v$, the cut $\delta V(J)$ consists of the edges from J to v and the terminals with a foot in J. So on $\mathcal{Z}(Q)$

the sum of the edges from J to v equals the sum of the terminals with a foot in J. Hence a is the sum of the terminals with a foot in Y. The sum of all terminals with a foot in K is zero on $\mathcal{Z}(Q)$, because these terminals form the cut $\delta V(K)$. As $a \neq 0$, some component of $K - v$ inside Y contains a foot, and some component outside Y contains a foot.

Step 2. In each component K of $Q - A$ fix a spanning tree. For each terminal add a new leaf at its foot. Let T_K be the union of the paths of this tree between the new leaves, and let T be the union of the trees T_K. Every leaf of T is a new leaf, so T has at most b leaves and at most b vertices of degree at least three. The exceptions are the relevant vertices that are feet or have degree at least three in T. There are at most $2b$ of them.

Let v be a relevant vertex that is not an exception. By Step 1 two components of $K-v$ contain feet. The path of the tree between the two new leaves at these feet passes through v. So v is a vertex of T, and its degree in T is two. A *thread* is a maximal path of T whose inner vertices have degree two in T. There are at most $2b$ threads, and every vertex of degree two of T is an inner vertex of exactly one thread. The strings are the sequences of the relevant vertices that are not exceptions on one thread, in the order of the thread.

Step 3. Let v be a vertex of a string, in the component K. The forest $T - v$ has exactly two components that meet K, one on each side of v along its thread. Their vertex sets in K are the *near side* of v, towards the start of the thread, and the *far side* of v. Each side is connected in $K-v$, and all feet in K lie in the two sides. By Step 1 the two sides lie in different components of $K - v$.

Now let v_i and v_j be as in the lemma. Let Π be the part of their thread strictly between them. It contains v_{i+1}, it is connected, and it contains no foot, because a new leaf at a vertex of Π would give that vertex degree three in T. Let $M(i, j)$ be the vertex set of the component of $K - v_i - v_j$ that contains Π. This gives (i).

For (ii), suppose that $M(i, j)$ contains a foot w. Then w is a vertex of T outside Π. So w lies in the near side of v_i or in the far side of v_j; say in the near side of v_i. The path Π lies in the far side of v_i. But w and Π are joined in $K - v_i$ through $M(i, j)$, and the two sides of v_i lie in different components of $K - v_i$. This is a contradiction. So $M(i, j)$ contains no foot. An edge that leaves $M(i, j)$ ends in A, in v_i or in v_j, and it does not end in A.

For (iii), let $j \leq i'$. The set $M(i, j)$ is connected in $K - v_j$ and contains Π, which lies in the near side of v_j. So $M(i, j)$ lies in the component of $K - v_j$ that contains the near side of v_j. We show that $M(i', j')$ lies in the component of $K - v_j$ that contains the far side of v_j; then the two sets are disjoint. The set $M(i', j')$ is connected in $K - v_{i'}$ and contains the part Π' of the thread between $v_{i'}$ and $v_{j'}$, which lies in the far side of $v_{i'}$. If $j = i'$, this is the claim. If $j < i'$, then v_j lies in the near side of $v_{i'}$, so $v_j \notin M(i', j')$. Hence $M(i', j')$ is connected in $K - v_j$, and it contains Π', which lies in the far side of v_j. ■

A packing bound.

Proposition 8.9 (Packing). Let G be a connected cubic multigraph. Let P be empty or induce a connected subgraph, and let S consist of s vertices outside P. Suppose $G - (P \cup S)$ has girth at least $D \geq D_*$ and $q(G, P \cup S) \geq \frac{2}{5}$. Let $2 \leq d \leq 2D$, and let $X_1, \dots, X_n$ be pairwise disjoint sets outside $P \cup S$, each inducing a connected graph with a cycle and satisfying $|\delta X_i| \leq d$. Then $n \leq 1551\, d(s+1)2^d$.

Proof. Contract each X_i to a marked vertex x_i, contract each component of $G - \bigcup_i X_i$ to a vertex, and delete the loops. The resulting graph Q is connected. Projection maps the uniform distribution on $\mathcal{Z}(G)$ to the uniform distribution on $\mathcal{Z}(Q)$, as in Lemma 2.6(b). Conditional on the projected edges, the restrictions to the induced graphs $G[X_i]$ are independent affine joins. Let I_i be the event that the projection uses no edge at x_i. On I_i the restriction to $G[X_i]$ is an even subgraph, and its forest probability is at most $1/2$, since $G[X_i]$ contains a cycle. Hence

$$\frac{2}{5} \leq q(G, P \cup S) \leq \mathbb{E} \prod_{i=1}^{n} \left(1 - \frac{\mathbf{1}_{I_i}}{2}\right). \tag{8.2}$$

Write $L_i = L_{x_i}$ for the local parity space in Q. The sum of its incident edge functionals is zero, so $\dim L_i \leq d - 1$ and $\mathbb{P}(I_i) = 2^{-\dim L_i} \geq 2^{1-d}$. If z_0 local spaces are zero, then $\frac{2}{5} \leq 2^{-z_0}$. Thus at most one local space is zero. Keep its vertex in Q, but omit this mark from the selection below. If no mark remains, the desired bound is immediate.

Choose a maximal direct-sum family A of nonzero local spaces. Put $t = |A| \geq 1$ and $b = |\delta A| \leq dt$. The events I_a for $a \in A$ are independent, so

$$\frac{2}{5} \leq \prod_{a \in A} \left(1 - \frac{\mathbb{P}(I_a)}{2}\right) \leq (1 - 2^{-d})^t < e^{-t2^{-d}}.$$

In particular, $t < 2^d$. If all remaining marks lie in A, the bound follows. Otherwise A is a nonempty proper vertex set of Q. Every remaining mark outside A is relevant in Lemma 8.8, by maximality. There are at most $2b$ exceptions and at most $2b$ strings.

Suppose a string has at least $(h_* + 2)(s + 3)$ marks. Split these marks into $s + 3$ consecutive groups of $h_* + 2$ marks. For each group take the region between its first and last marks from Lemma 8.8. Lift this region to the union of the sets contracted to its vertices. The lifted regions are connected and pairwise disjoint. Each contains h_* disjoint cyclic sets X_i, so has rank at least h_*. Its boundary has at most $2d \leq 4D$ edges. At most s regions meet S, and at most one meets P, since a nonempty connected P lies in one contracted set. Thus two regions remain inside $G - (P \cup S)$. Proposition 8.7(a) gives $q(G, P \cup S) \leq \frac{9}{25} < \frac{2}{5}$, a contradiction.

Count the omitted mark, the selected marks, the exceptions and the strings:

$$n \leq 1 + t + 2b + 2b(h_* + 2)(s + 3).$$

Here $b \leq dt$, $t < 2^d$, $s + 3 \leq 3(s + 1)$ and $h_* + 2 = 258$. So $n \leq 2^d + 2d\, 2^d + 1548\, d(s+1)2^d \leq 1551\, d(s+1)2^d$. ∎

Raising the girth.

Theorem 8.10 (Raising the girth). Let G be a connected cubic multigraph and let P be empty or induce a connected subgraph. Suppose $q(G,P) \geq \frac{2}{5}$ and $G - P$ has girth at least D_*. Put $D_j = 2^j D_*$ for $j \geq 0$. There are nested sets $S_0 \subseteq S_1 \subseteq \cdots$ outside P such that

(i) S_j is the union of a maximal family of pairwise disjoint cycles of length at most D_j in $G - P$; hence $G - (P \cup S_j)$ has girth greater than D_j;

(ii) $\log(|S_j| + 1) \leq 2D_j + 6\log^2 D_j$.

Proof. Put $S_{-1} = \emptyset$. Having chosen S_{j-1}, take a maximal disjoint family $\mathcal{Q}_j$ of cycles of length at most $b = D_j$ in $G - (P \cup S_{j-1})$, and add its vertices to S_{j-1} to form S_j. The exterior girth is at least $a = D_*$ for $j = 0$, and at least $a = D_{j-1} + 1$ otherwise; thus $b \leq 2a$. Each cycle C has $|\delta V(C)| \leq |C| \leq b$. Also $q(G, P \cup S_{j-1}) \geq q(G,P) \geq \frac{2}{5}$. Proposition 8.9 gives

$$|\mathcal{Q}_j| \leq 1551\, b(|S_{j-1}| + 1)2^b, \qquad |S_j| + 1 \leq 1552\, b^2 2^b (|S_{j-1}| + 1).$$

The union of all families chosen so far is maximal at scale D_j, which proves (i). For (ii), take logarithms in the recursion. As $\log 1552 < 11$, the step i adds at most $D_i + 2\log D_i + 11$. There are $j + 1 \leq \log D_j$ steps, and $\sum_{i \leq j} D_i < 2D_j$. So

$$\log(|S_j| + 1) \leq 2D_j + \log D_j \left(2\log D_j + 11\right) \leq 2D_j + 6\log^2 D_j,$$

because $\log D_j \geq \log 7 > \frac{11}{4}$. ∎

8.3 Links, a second moment and a chain

Components between two connected sets. Lemma 2.5 bounds the number of disjoint cyclic regions with small boundaries, and it needs an excess above $\frac{1}{2}$. For components that join the same two connected sets there is a bound without an excess.

Lemma 8.11 (Parallel links). Let Γ be a connected multigraph, let A and B be disjoint regions, and let P be a set of vertices with $q(\Gamma, P) \geq \frac{2}{5}$. Let $t \geq 0$ be an integer. Then at most $5 \cdot 2^t$ components Y of $\Gamma - (A \cup B)$ have all of the following properties: Y has an edge to A and an edge to B; Y avoids P; Y contains a cycle; $|\delta Y| \leq t$.

Proof. Let $Y_0, Y_1, \ldots, Y_n$ be such components. For $1 \leq i < j \leq n$ the graph $\Gamma - Y_i - Y_j$ is connected. Indeed, it contains the connected set $A \cup B \cup Y_0$, and every other component of $\Gamma - (A \cup B)$ is connected and has an edge to $A \cup B$. No edge joins Y_i and Y_j. So the regions Y_i and Y_j have exactly one link, and by Lemma 2.3

$$\mathbb{P}(\mathcal{O}_{Y_i} \cap \mathcal{O}_{Y_j}) = \mathbb{P}(\mathcal{O}_{Y_i})\, \mathbb{P}(\mathcal{O}_{Y_j}).$$

Let E_i be the event that $\mathcal{O}_{Y_i}$ holds and $\xi \cap E(\Gamma[Y_i])$ is not empty. By the last part of Lemma 2.3, the events E_i and E_j are independent as well, and $\mathbb{P}(E_i) \geq 2^{-t} \cdot \frac{1}{2}$. If $n = 0$, the bound holds.

Otherwise let N be the number of the events $E_1, \dots, E_n$ that occur, and put $\nu = \mathbb{E}N \geq n\,2^{-t-1} > 0$. Then $\mathbb{E}N^2 \leq \nu^2 + \nu$, and the Cauchy–Schwarz inequality gives

$$\mathbb{P}(N = 0) \;\leq\; 1 - \frac{\nu^2}{\mathbb{E}N^2} \;\leq\; \frac{1}{1+\nu}.$$

If $\xi \cap E(\Gamma - P)$ is a forest, then no E_i occurs, because Y_i avoids P and a nonempty even subgraph is not a forest. So $\frac{2}{5} \leq q(\Gamma, P) \leq 1/(1+\nu)$. Hence $\nu \leq \frac{3}{2}$ and $n \leq 3 \cdot 2^t$. With Y_0 there are at most $3 \cdot 2^t + 1$ components. ∎

Lemma 8.12 (Few links). Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and $q(G, P) \geq \frac{2}{5}$. Let S be the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D in $G - P$, and put $J = G - (P \cup S)$. Let $8 \leq D \leq L$, and let t be an integer with $1 \leq t \leq D/4$. Then any two disjoint cycles C, C' of J of length at most L satisfy

$$\ell\big(V(C), V(C')\big) \;\leq\; 1 + \frac{2L}{t+1} + 5 \cdot 2^t + 289\Big(\frac{L}{D}\Big)^2.$$

Proof. By the maximality of the family, J is a simple graph of girth greater than D. So we are in the setting of Section 3.2, with $\Gamma = G$ and $U = P \cup S$. Put $U' = V(C) \cup V(C')$. Every vertex of U' lies on exactly one edge outside its own cycle, so at most $2L$ edges leave U'. At most one component of $G - U'$ meets P, because $G[P]$ is connected and disjoint from U'. At most $2L/(t+1)$ components have more than t edges to U'. Of the other components that are joined to both cycles, at most $5 \cdot 2^t$ contain a cycle, by Lemma 8.11 with $A = V(C)$ and $B = V(C')$. The remaining components that are joined to both cycles induce trees, avoid P and have at most t edges to U'. They also avoid S: each cycle of the family is disjoint from U', so it lies in one component of $G - U'$, and this component contains a cycle. By Lemma 3.7, at most $289(L/D)^2$ links are such components or edges. ∎

Pairs with exactly two links. Let G, P, S, J, D and L be as in Lemma 8.12. As in its proof we are in the setting of Section 3.2. So $s = \lfloor (D-1)/2 \rfloor$, the set $\mathcal{C}$ consists of the cycles of J of length at most L, and the numbers a_C, μ and W_2 and the random variables X_C and Z are defined there.

Lemma 8.13 (Two links). Suppose that $\ell(V(C), V(C')) \leq b$ for all disjoint $C, C' \in \mathcal{C}$, and put $\eta = 2^b L^3 2^{-s}$. Then

$$W_2 \;\geq\; \tfrac{2}{3}\,\mu^2 - (1+\eta)\,\mu.$$

Proof. By Lemma 3.6, $\mathbb{E}Z^2 \leq \mu^2 + W_2 + (1+\eta)\mu$. If $\xi \cap E(G-P)$ is a forest, then ξ contains no cycle of J, so $Z = 0$. Hence $\mathbb{P}(Z \neq 0) \leq \frac{3}{5}$. The Cauchy–Schwarz inequality $\mu^2 \leq \mathbb{E}Z^2 \cdot \mathbb{P}(Z \neq 0)$ gives $\mathbb{E}Z^2 \geq \frac{5}{3}\mu^2$. ∎

IDEA | from pairs with two links to a chain

By Lemma 8.13 there are many pairs of cycles with exactly two links. Let C and C' be such a pair. One of the two links avoids p, and then C' separates a neighbour u of C from p. Many disjoint cycles C' have this property for the same C and u. They are nested (Figure 8), and the parts between them are the pieces that we need. The argument requires that C' lies in the component of p in $G - V(C)$. By symmetry, at least half of the weight of the pairs with two links has this property.

Separators are nested.

Lemma 8.14 (Nested separators). Let u and p be two vertices of a connected graph H. Let C_1 and C_2 be disjoint connected sets of vertices that avoid u and p, such that u and p lie in different components of $H - C_i$ for $i = 1, 2$. Let U_i be the vertex set of the component of u in $H - C_i$. Then exactly one of the following holds: $C_1 \subseteq U_2$ and $U_1 \subseteq U_2$; or $C_2 \subseteq U_1$ and $U_2 \subseteq U_1$.

Proof. Take a path from u to p, and let z be its last vertex before the first vertex in C_1. The part of the path up to z lies in U_1. In the same way there is a path from p to a vertex next to C_1 that avoids C_1. The set C_2 is connected and disjoint from C_1, so it lies in one component of $H - C_1$. If this component is neither U_1 nor the component of p, then the two partial paths and a path inside C_1 form a walk from u to p that avoids C_2. This is impossible. So C_2 lies in U_1 or in the component of p in $H - C_1$, and the same holds with the indices exchanged.

Suppose that C_2 lies in the component of p in $H - C_1$. Then C_2 avoids U_1. So every vertex of U_1 is joined to u outside C_2, and so is every vertex of C_1, through z and the connected set C_1. Hence $U_1 \cup C_1 \subseteq U_2$. If C_2 lies in U_1 and C_1 lies in U_2, then the path from p to a vertex next to C_1, a path inside C_1, and a path inside U_2 from C_1 to u give a walk from p to u outside C_2. This is impossible. So if C_2 lies in U_1, then C_1 lies in the component of p in $H - C_2$, and $U_2 \cup C_2 \subseteq U_1$ by the first case with the indices exchanged. The two alternatives exclude each other, because $C_1 \subseteq U_2$ and $U_2 \subseteq U_1$ would give $C_1 \subseteq U_1$. ∎

So a family of pairwise disjoint connected sets that separate u from p is linearly ordered: write $C_1 < C_2$ in the first case of the lemma. The order is transitive, because it is the order of the sets U_i by inclusion.

Lemma 8.15 (Blocks). In the situation of Lemma 8.14, let $C_1 < C_2$ and put $X = U_2 \setminus (U_1 \cup C_1)$. Then every edge of H that leaves X ends in C_1 or in C_2. If $C_1 < C < C_2$ for a third set C of the family, then $C \subseteq X$. If $C_1 < C_2 \leq C_3 < C_4$, then the sets $U_2 \setminus (U_1 \cup C_1)$ and $U_4 \setminus (U_3 \cup C_3)$ are disjoint.

Proof. An edge that leaves U_2 ends in C_2, because U_2 is a component of $H - C_2$. No edge joins X to U_1: its end in X would lie in the component U_1 of $H - C_1$. So an edge from X to $U_1 \cup C_1$ ends in C_1. For the second statement, $C \subseteq U_2$, and C avoids $U_1 \cup C_1$ because $U_1 \cup C_1 \subseteq U_C$, the component of u in $H - C$. For the third, $U_2 \subseteq U_3$. ∎

The chain.

Lemma 8.16 (Chain). In the situation of Lemma 8.13, suppose that

$$\mu > 30, \qquad \eta \le \frac{1}{2}, \qquad L^2 2^{-s} \le \frac{1}{2}.$$

Fix a vertex $p \in P$. Then there are a cycle $C \in \mathcal{C}$, a vertex $u \notin V(C)$ that has a neighbour on C, and pairwise disjoint cycles $C_1, \dots, C_T \in \mathcal{C}$ with

$$T \ \ge \ \frac{8 \cdot 2^s}{L^3},$$

such that each C_i is disjoint from C and avoids u, and u and p lie in the same component of $G - V(C)$ and in different components of $G - V(C) - V(C_i)$.

Proof. For disjoint $C, C' \in \mathcal{C}$ say that C' is *far behind* C if C' does not lie in the component of p in $G - V(C)$. This relation is antisymmetric. Let C' be far behind C. A path from p to its first vertex on C stays in the component of p in $G - V(C)$ before it ends. So it avoids C'. As C is connected and disjoint from C', all of C lies in the component of p in $G - V(C')$. So C is not far behind C'. Hence the ordered pairs (C, C') with C' far behind C have total weight at most $W_2/2$ when restricted to the exactly-two-link pairs. Indeed, the two-link relation is symmetric and opposite orientations have the same weight; at most one orientation is far behind.

Fix C. Every vertex of C has exactly one edge end outside C, so at most L vertices outside C have a neighbour on C. A cycle C' that is joined to C by an edge passes through one of them. By (3.2) these cycles have total weight at most $L^2 2^{-s} \le \frac{1}{2}$. So the ordered pairs of disjoint cycles that are joined by an edge have total weight at most $\mu/2$.

Call an ordered pair (C, C') *good* if the cycles are disjoint, have exactly two links, are not joined by an edge, and C' is not far behind C. By Lemma 8.13 and the two bounds above, the good pairs have total weight at least $W_2/2 - \mu/2$, hence at least

$$\frac{1}{3}\mu^2 - \left(1 + \frac{\eta}{2}\right)\mu \ > \ \left(10 - \frac{5}{4}\right)\mu \ > \ 8\mu.$$

So for some $C \in \mathcal{C}$ the cycles C' with (C, C') good have total weight more than 8. Fix this C, and put $H = G - V(C)$.

Let (C, C') be good. The two links are components Y_1, Y_2 of $G - V(C) - V(C')$ that are joined to both cycles, because no edge joins the cycles. The vertex p lies in at most one of them. So one of them, say Y, avoids p. Let u be a vertex of Y with a neighbour on C. The cycle C' lies

in the component of p in H, and Y is connected and joined to C'. So u lies in that component as well. The set Y is a component of $H - V(C')$ that contains u and not p. So C' separates u from p in H.

There are at most L choices for u. So for some vertex u the cycles C' that separate u from p in H, as above, have total weight at least $8/L$. Take a maximal family $C_1, \dots, C_T$ of pairwise disjoint cycles among them. Every cycle of the collection meets one of the at most TL vertices of the family. By (3.2),

$$\frac{8}{L} \;\le\; TL \cdot L\, 2^{-s}. \qquad \blacksquare$$

8.4 Proof of Theorem 8.1

Let D_* and C_0 be as in Lemma 8.4. Let G and P be as in the theorem, and suppose that $q(G,P) \ge \frac{2}{5}$. We derive a contradiction when r is large. All limits below are for $r \to \infty$, and they are uniform in G and P. Put

$$x = \log r, \qquad z = \log x, \qquad A = 6\, e^{124}, \qquad L = \lfloor Ax \rfloor, \qquad t = \lfloor z - 2\log z \rfloor.$$

Two girth scales. Let $D_j = 2^j D_*$ and let $S_0 \subseteq S_1 \subseteq \cdots$ be the sets of Theorem 8.10. Let D^- be the first D_j that is at least $\sqrt{x}$, and let D^+ be the first D_j that is at least $x/16$. For large r,

$$\sqrt{x} \le D^- < 2\sqrt{x}, \qquad \frac{x}{16} \le D^+ < \frac{x}{8}, \qquad D^- < D^+.$$

Let $S^- \subseteq S^+$ be the corresponding sets, and put $J^\pm = G - (P \cup S^\pm)$. Then J^- has girth greater than D^-, J^+ has girth greater than D^+, and $J^+ \subseteq J^-$. By Theorem 8.10(ii),

$$|S^-| \;\le\; 2^{4\sqrt{x}+6\log^2(2\sqrt{x})} = r^{o(1)}, \qquad |S^+| \;\le\; 2^{x/4+6\log^2 x} = r^{1/4+o(1)}. \tag{8.3}$$

We use the small scale D^- for the girth of the final pieces, and the large scale D^+ for a long chain.

Cycles at the large scale. We apply Section 8.3 with $S = S^+, J = J^+, D = D^+$ and $s = \lfloor (D^+ - 1)/2 \rfloor \ge x/32 - 1$. The hypotheses of Lemma 8.12 hold for large r: $8 \le D^+ \le L$ and $1 \le t \le z \le D^+/4$. So every two disjoint cycles of J^+ of length at most L have at most

$$b \;=\; 1 + \frac{2L}{t+1} + 5 \cdot 2^t + 289\Big(\frac{L}{D^+}\Big)^2 \;\le\; 1 + \frac{2Ax}{z - 2\log z} + \frac{5x}{z^2} + 289\,(16A)^2 \;\le\; \frac{3Ax}{z}$$

links. Hence

$$\eta = 2^b L^3 2^{-s} \;\le\; r^{3A/z}\,(Ax)^3\, 2\, r^{-1/32} = r^{-1/32+o(1)} = o(1), \qquad L^2 2^{-s} = o(1).$$

The graph J^+ has $v = 2r - 2 - |P| - |S^+| = 2r - o(r)$ vertices, and its deficit is $d_J \le 3(|P| + |S^+|) = r^{1/4+o(1)}$. Let μ be the total weight of its cycles of length at most L. By Lemma 3.5 with $\ell_0 = \lceil 6x \rceil$,

$$\Big(1 + \tfrac{3}{2} L^2\, 2^{-s}\Big)\,\mu \;\ge\; \Big(1 - \frac{L\, d_J}{2v} - 3v\, r^{-3}\Big) \sum_{\ell} \frac{1}{2\ell},$$

where the sum is over the even ℓ with $6x \le \ell \le L$. This sum is $(\frac{1}{4} - o(1))\ln(A/6) = 31 - o(1)$, and the two factors in brackets are $1 + o(1)$. Thus $\mu > 30$ for large r.

The chain. Fix $p \in P$. For large r the hypotheses of Lemma 8.16 hold. So there are a cycle C of J^+ of length at most L, a vertex u, and pairwise disjoint cycles $C_1, \dots, C_T$ of J^+ of length at most L with

$$T \;\geq\; \frac{8 \cdot 2^s}{L^3} \;\geq\; \frac{4}{A^3} \cdot \frac{r^{1/32}}{x^3},$$

such that each C_i separates u from p in $G - V(C)$. Let H be the component of $G - V(C)$ that contains u and p. Each $V(C_i)$ is connected and separates u from p, so it lies in H. By Lemma 8.14 we may number the cycles so that $C_1 < C_2 < \cdots < C_T$. Let U_i be the vertex set of the component of u in $H - V(C_i)$.

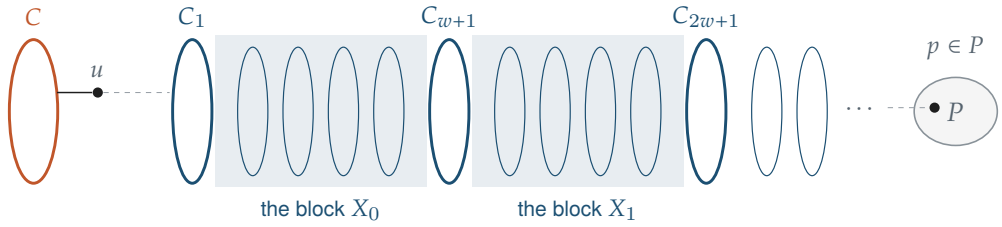


Figure 8. The chain of Lemma 8.16. Each cycle C_i separates u from p in $G - V(C)$, so the cycles are nested. Every w-th cycle is the end of a block. A block has many vertices and at most $2L$ boundary edges.

Blocks. Put $w = \lceil 128A^2x \rceil + 2$. For $l = 0, 1, \dots$ with $(l+1)w \leq T - 1$ let

$$X_l \;=\; U_{(l+1)w+1} \setminus \big(U_{lw+1} \cup V(C_{lw+1})\big).$$

There are at least $\lfloor (T-1)/w \rfloor \geq r^{1/32}/x^5$ such sets, for large r. By Lemma 8.15 they are pairwise disjoint. Each contains the $w - 1$ cycles between its two end cycles. These cycles are pairwise disjoint and each has more than D^+ vertices, so

$$|X_l| \;\geq\; (w-1)(D^+ + 1) \;\geq\; 128A^2x \cdot \frac{x}{16} = 8A^2x^2 \;\geq\; 8L^2.$$

Every edge of H that leaves X_l ends on one of its two end cycles. An end cycle has at most L vertices, and each of them has one edge end outside the cycle. So at most $2L$ edges of H leave X_l.

Clean blocks. Discard a set X_l if it contains a vertex with a neighbour on C, or a vertex of S^-. At most L vertices outside C have a neighbour on C, so this discards at most $L + |S^-| = r^{o(1)}$ sets. For large r at least two sets remain. Let X be one of them.

- X avoids P. The set P is connected and disjoint from the cycles, and it contains p, which is not in U_i for any i. A path in $G[P]$ from p to a vertex of U_i would have to pass through C_i or through C.
- No edge of G joins X to $V(C)$, and H is a component of $G - V(C)$. So the edges of G that leave X are edges of H, and $|\delta X| \leq 2L$.
- X avoids $P \cup S^-$, so $G[X]$ is a subgraph of J^-, and its girth is greater than D^-.

The contradiction. Two of the remaining sets satisfy the hypotheses of Proposition 8.7(b) with $\Lambda = L$ and $D = D^-$: they are disjoint, they avoid P, each has at least $8L^2$ vertices and at most

$2L$ boundary edges, and $D^- \geq \sqrt{x} \geq C_0 \log(L+2)$ for large r. Thus $q(G,P) \leq \frac{9}{25} < \frac{2}{5}$. This contradiction proves Theorem 8.1. ∎

9 Proof of Theorem 1.1

Let D_* be as in Theorem 8.1 and K_0 as in Section 5. By (1.1),

$$H(r) \;\geq\; 2^{r-1} + (r-2)\,2^{r/2-1} \qquad \text{for all } r \geq 3. \tag{9.1}$$

Choose $R \geq 8$ so large that the following holds for every $r \geq R$: Proposition 6.3, Theorem 6.8, Theorem 7.1 with $g_0 = D_* - 1$, and Theorem 8.1 apply to the rank r; and $r - 2 > 10K_0^2$ and $2\log(2r) + 2 \leq 3\log r$.

Let $r \geq R$. A ring R_r has $H(r)$ cycles by Corollary 6.2, so $\psi(r) \geq H(r)$. By Lemma 3.10, some connected cubic multigraph G of rank r has $\psi(r)$ cycles. We show that G is a ring or that $c(G) \leq 2^{r-1} + 5K_0^2\,2^{r/2}$. In the first case $c(G) \leq H(r)$ by Corollary 6.2. In the second case $c(G) < H(r)$ by (9.1), because $r - 2 > 10K_0^2$. So $\psi(r) \leq H(r)$.

Case 1: G has a loop e. Every cycle of G other than e is a cycle of $G - e$, which has rank $r-1$. So $c(G) \leq 1 + (2^{r-1} - 1)$.

Case 2: G has no loop, and it has a cyclic cut of size at most three. Then $c(G) \leq 2^{r-1} + 5K_0^2\,2^{r/2}$ by Proposition 6.3.

Case 3: G has no loop, it has a cyclic cut of size four, and it has no smaller cyclic cut. Then G is a ring or $c(G) < 2^{r-1}$, by Theorem 6.8.

In the other cases G has no loop and no cyclic cut of size at most four. Let g be its girth.

Case 4: $g < D_*$. Then $c(G) < 2^{r-1}$ by Theorem 7.1.

Case 5: $g \geq D_*$. Let W be a shortest cycle, and put $P = V(W)$. As in the proof of Corollary 3.11, $|P| = g \leq 2\log(2r) + 2 \leq 3\log r$. The graph $G[P]$ is connected, and $G - P$ has girth at least D_*. So $q(G,P) < \frac{2}{5}$ by Theorem 8.1.

Let Z be a cycle of G that meets P. Its edges in $G - P$ form a proper subset of Z, hence a forest. So at most $q(G,P)\,2^r < \frac{2}{5}\,2^r$ cycles meet P. The other cycles are cycles of $G - P$, and there are at most $2^{\beta(G-P)}$ of them. The cycle W has no chord. So $G - P$ arises from G by the deletion of g vertices and $2g$ edges, and

$$\beta(G-P) = r - 1 - g + \mathrm{comp}(G-P).$$

The graph G has no cut of size one, because such a cut is cyclic (Section 6). So every component of $G - P$ contains the ends of at least two of the g edges that leave P. Hence $\mathrm{comp}(G-P) \leq g/2$ and $\beta(G-P) \leq r - 1 - g/2 \leq r - 4$, because $g \geq 7$. Therefore $c(G) < \left(\frac{2}{5} + \frac{1}{16}\right)2^r < 2^{r-1}$. ∎

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